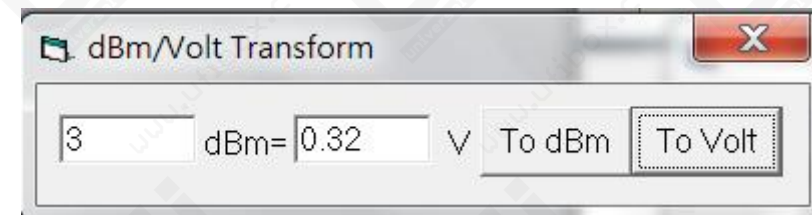
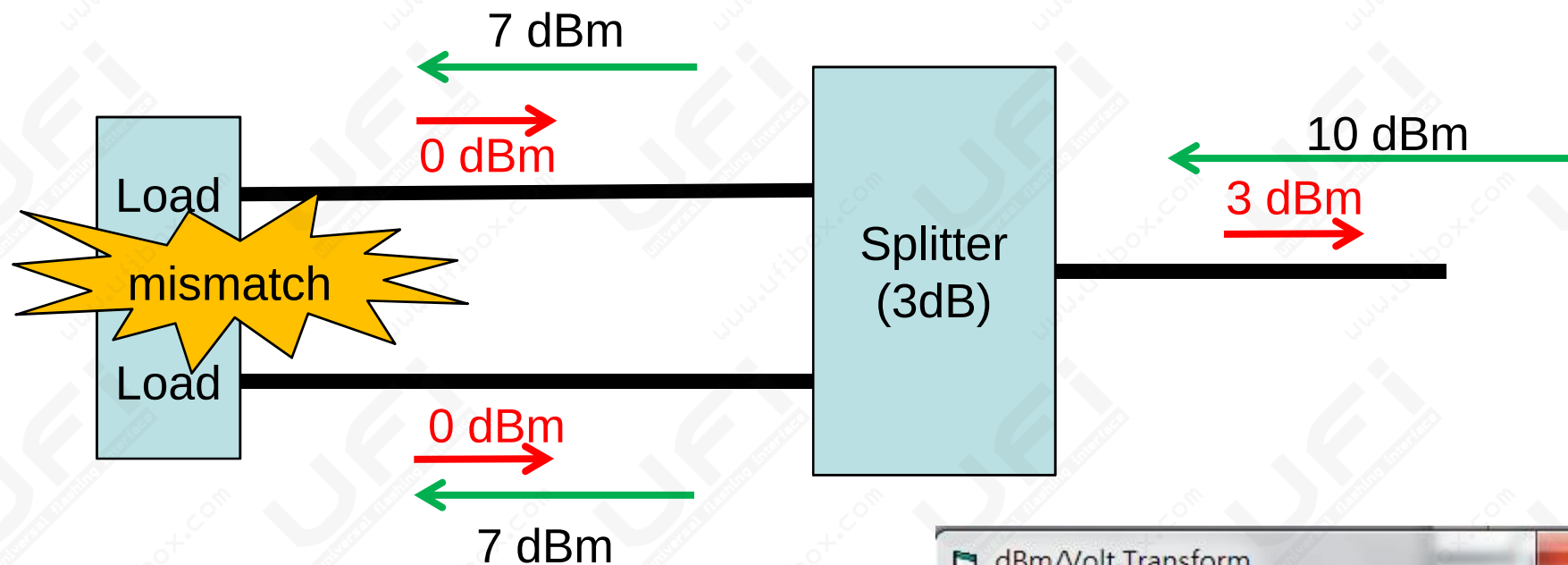
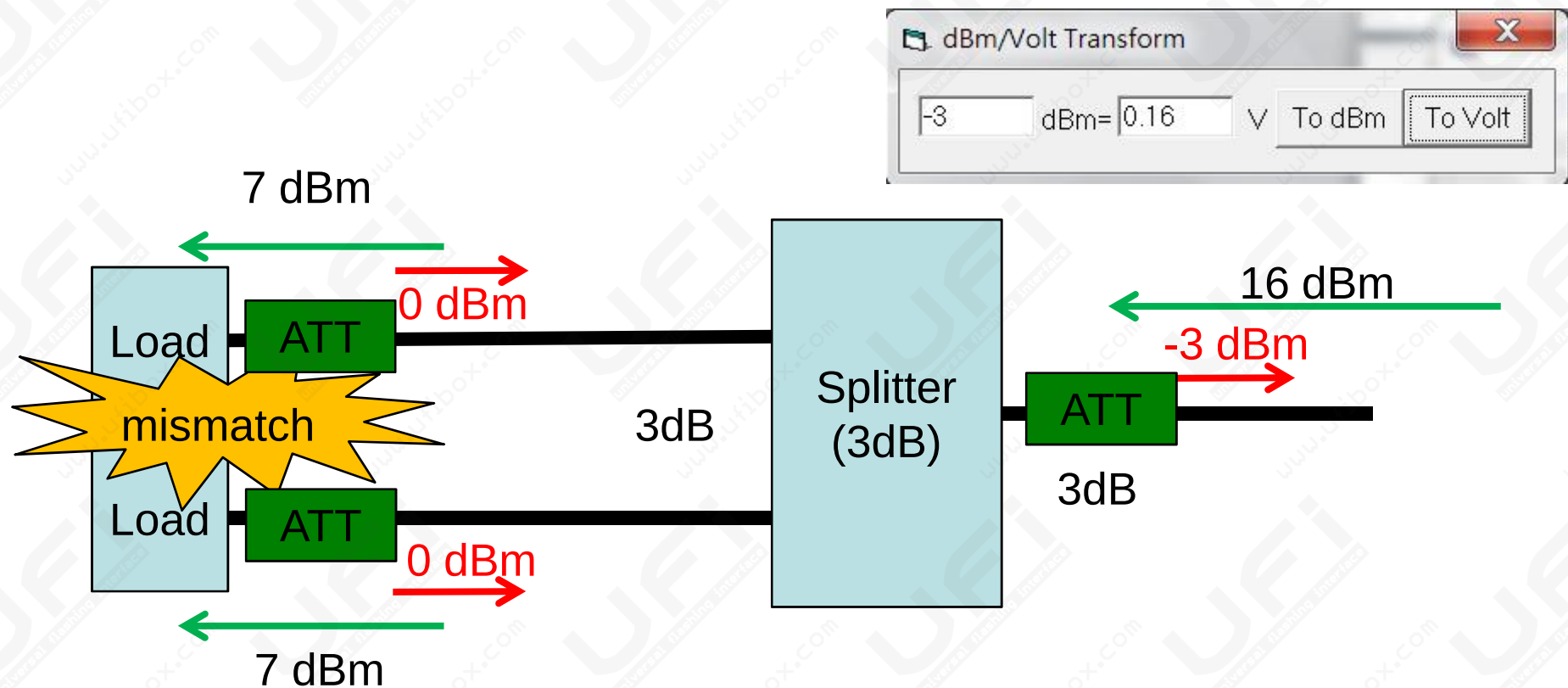

ASUS ZE550KL&551KL

RF Hardware Trouble Shooting Guide

How to Solve RF System Mismatch Issue (1/2)



How to Solve RF System Mismatch Issue (2/2)



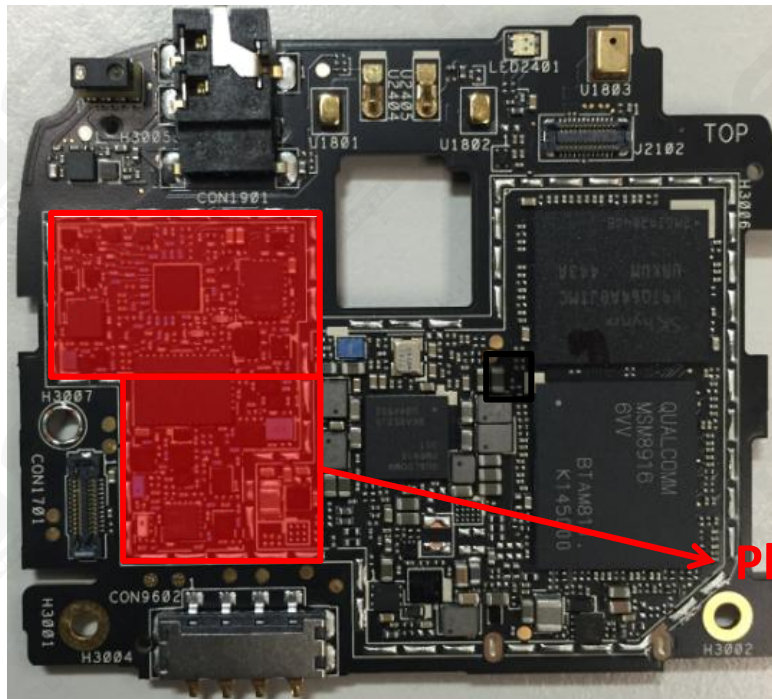
Use the combiner connection single port of instrument , as shown to make up the loss, at least 9dB.

Splitter Diff. output port improvement 6dB isolation.

Currently to test the run using 2 port connection test, but we need to add 3dB attenuator

ZE550KL RF Areas

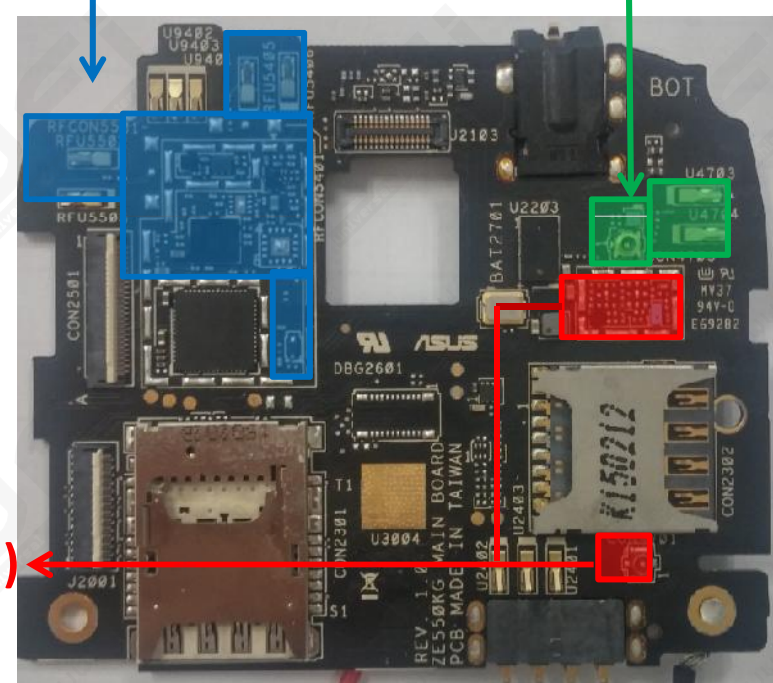
Top side ↓



Bottom side ↓

BT/Wi-Fi/GPS

Phone (Diversity)



Phone (2/3/4 G)

ZE550KL RF Supported Bands

- *All SKU Supported Bands

- 2G: EDGE/GSM : 2/3/5/8

- 3G: WCDMA :1/2/5/8

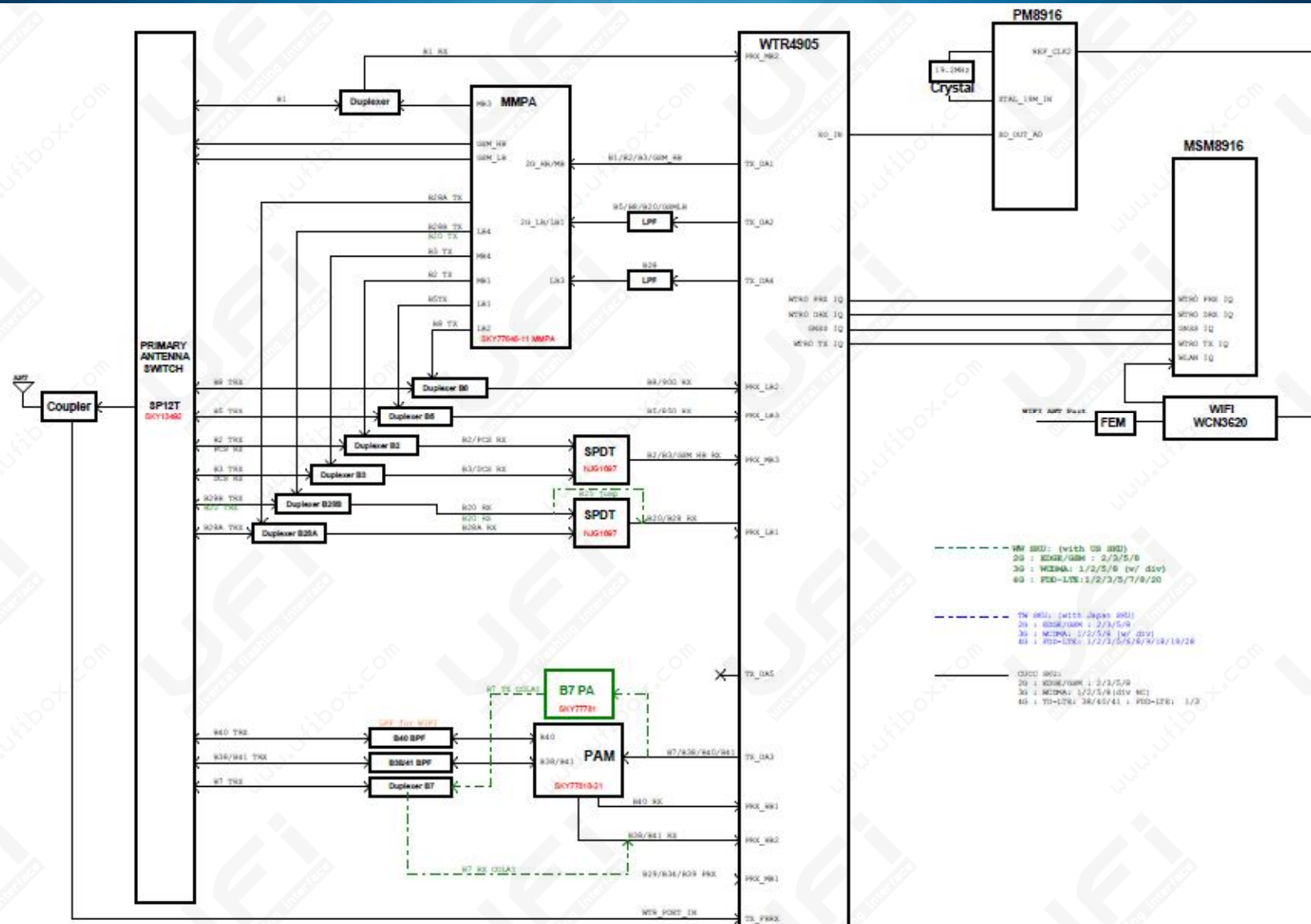
- 3G: WCDMA Div. :1/2/5/8 (WW&TW) ; 1 (CUCC)

- * WW SKU: LTE FDD Bands 1/2/3/5/7/8/20

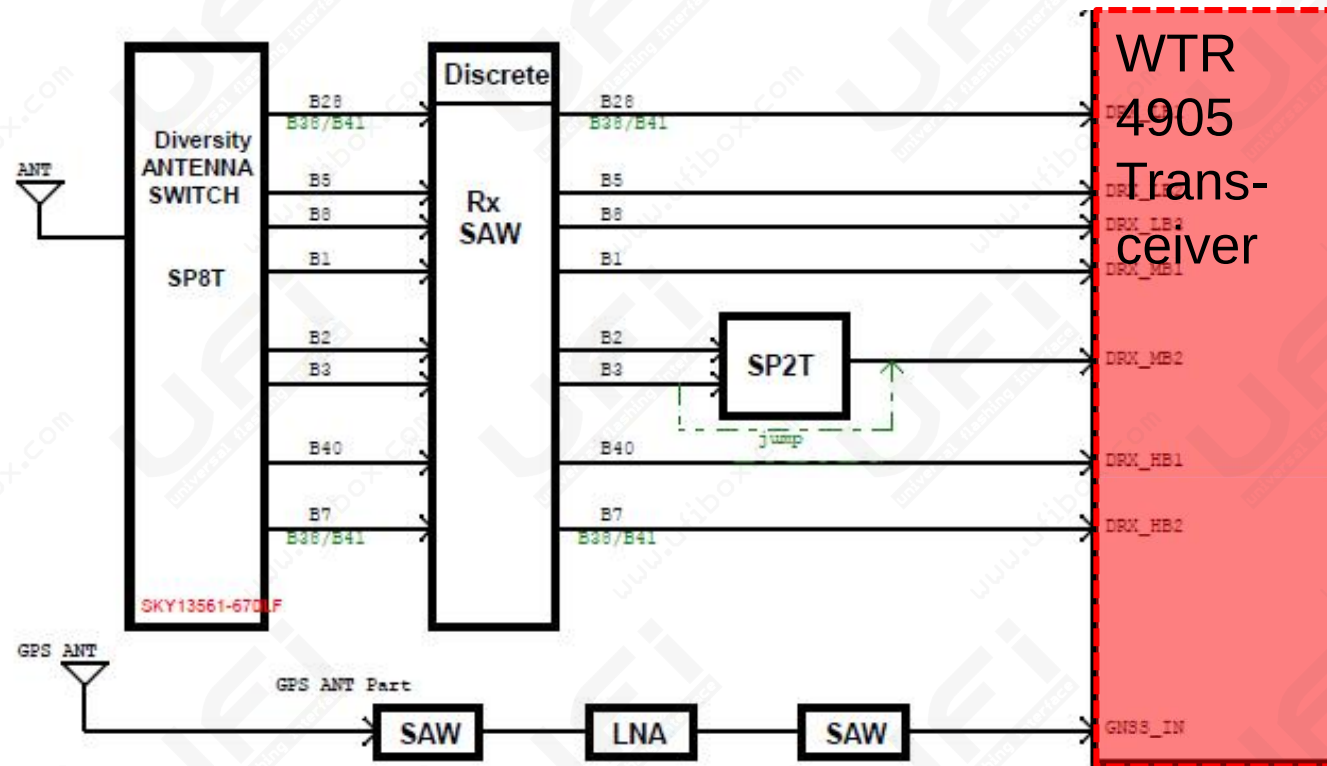
- * TW SKU: LTE FDD Bands 1/2/3/5/6/8/9/18/19/28

- * CUCC SKU: LTE FDD Bands 1/3; TDD Bands 38/40/41

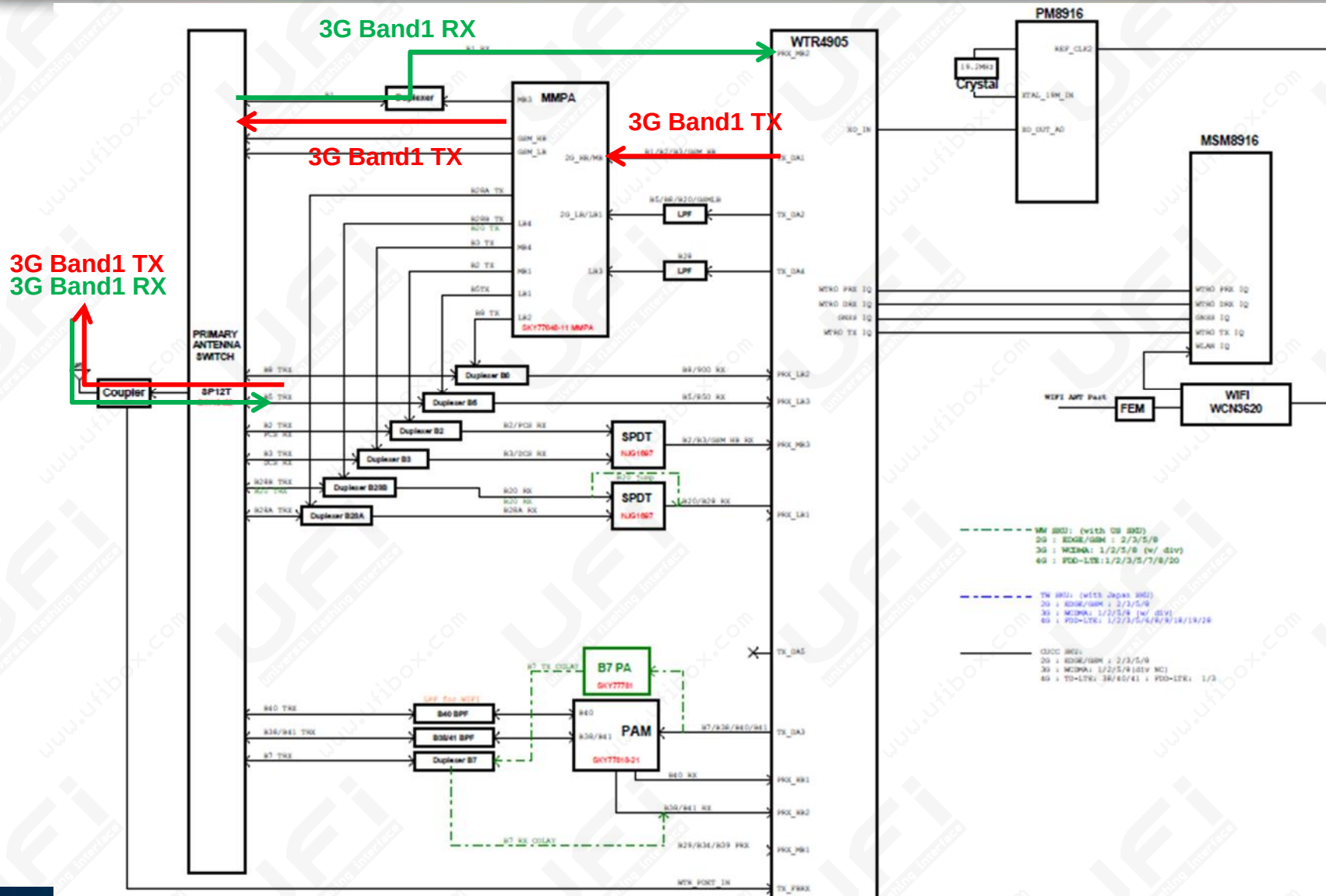
RF Block Diagram : 2G/3G/4G System (1)



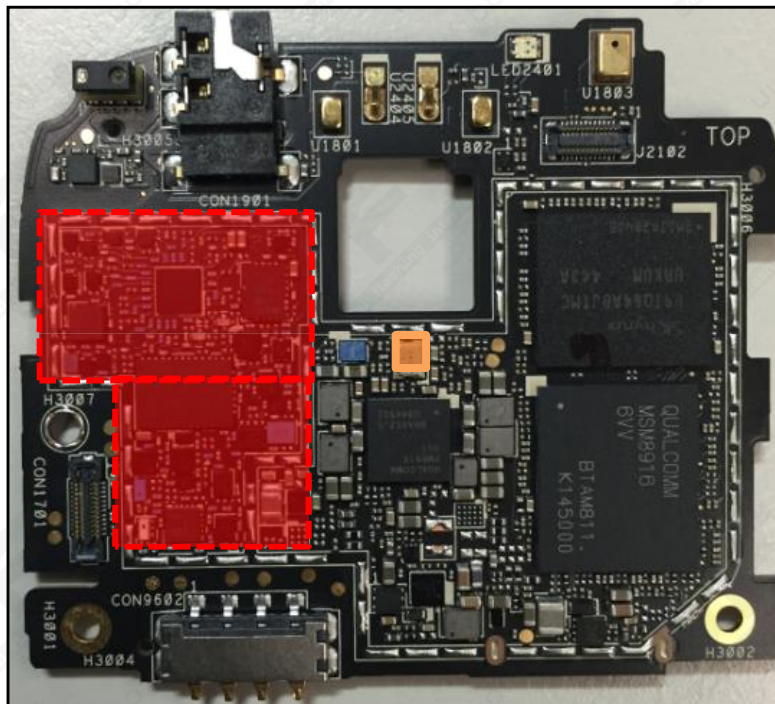
RF Block Diagram : 2G/3G/4G System (2)



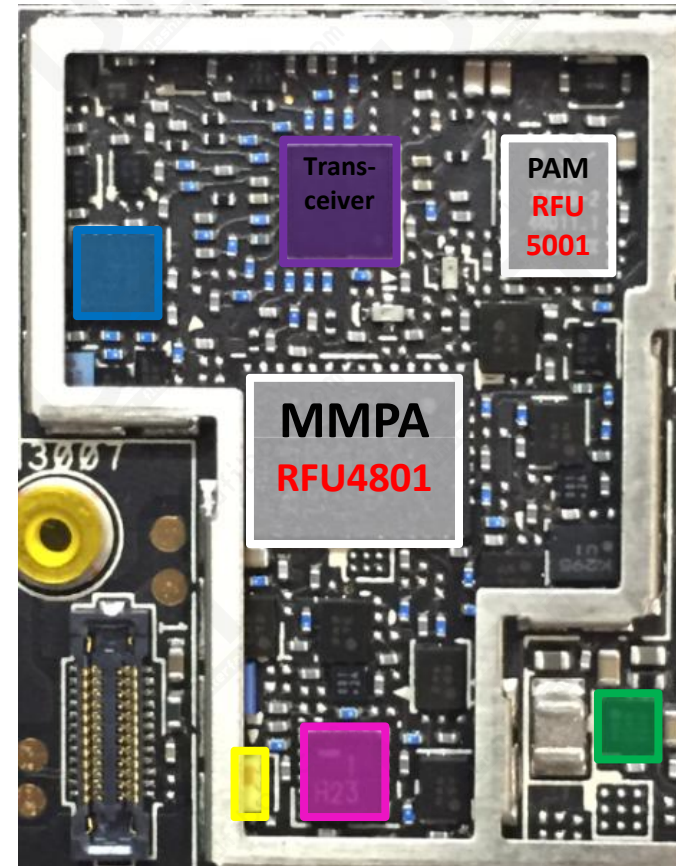
RF Block Diagram : 2G/3G/4G System (Ex : 3G Band1)



Key Components : Phone (1/2)



 19.2MHz Crystal **X8501**



 Transceiver **RFU4501**

Trans-
ceiver

PAM
RFU
5001

MMPA
RFU4801

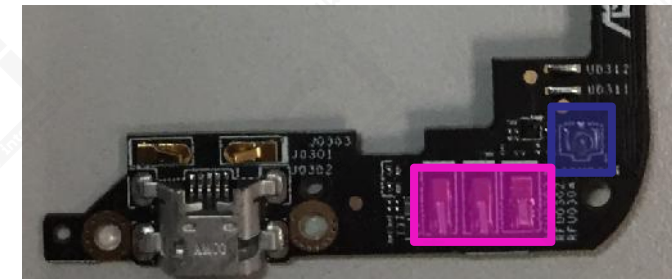
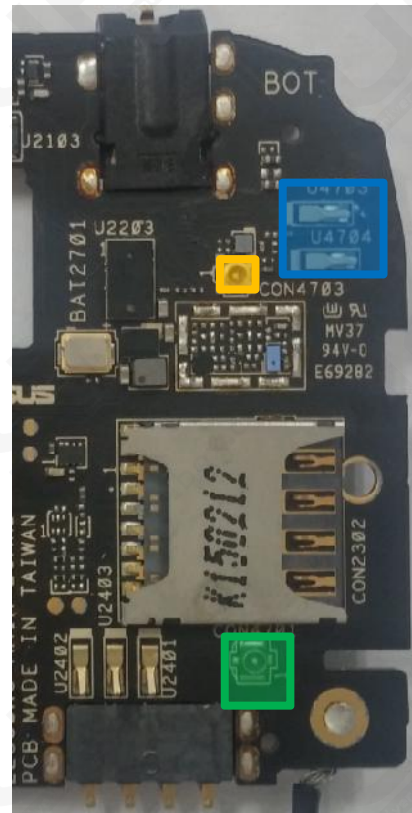
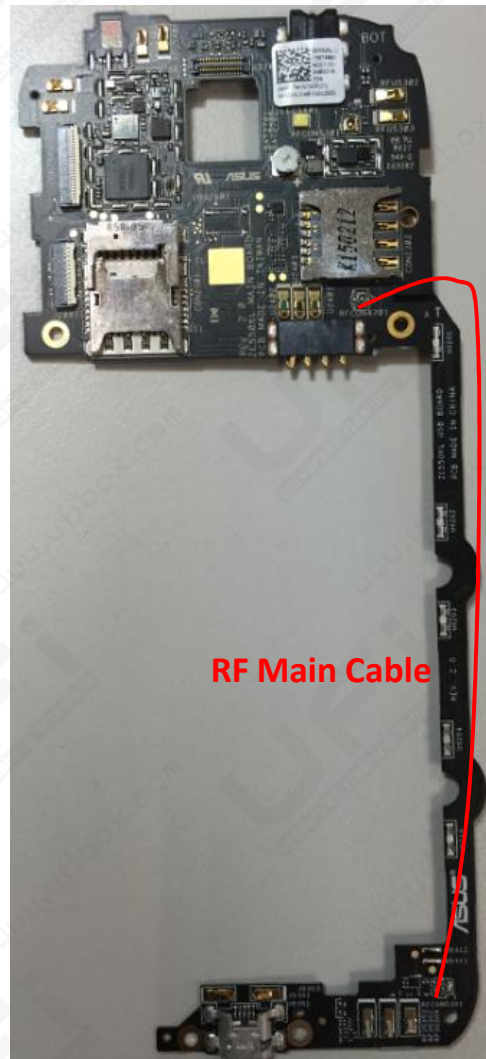
 DCDC **RFU4601**

 Diversity Antenna SW **RFU5301**

 Coupler **RFU4702**

 Main Antenna SW **RFU4701**

Key Components : Phone (2/2)



Spring (Phone Div. antenna)



RF Main Connector **RFCON4701**



RF Diversity Connector
RFCON5301

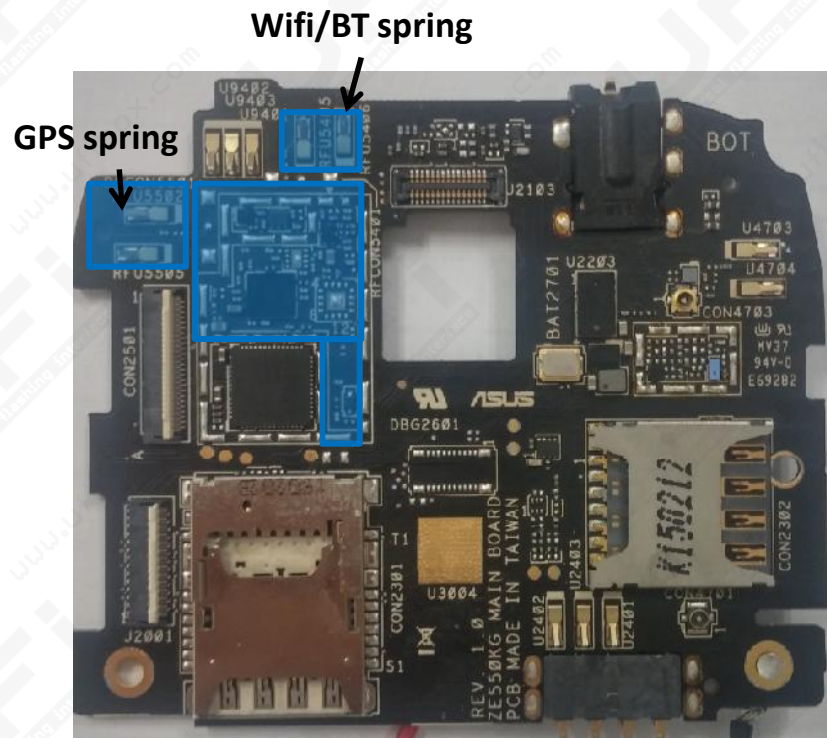


RF Main Connector

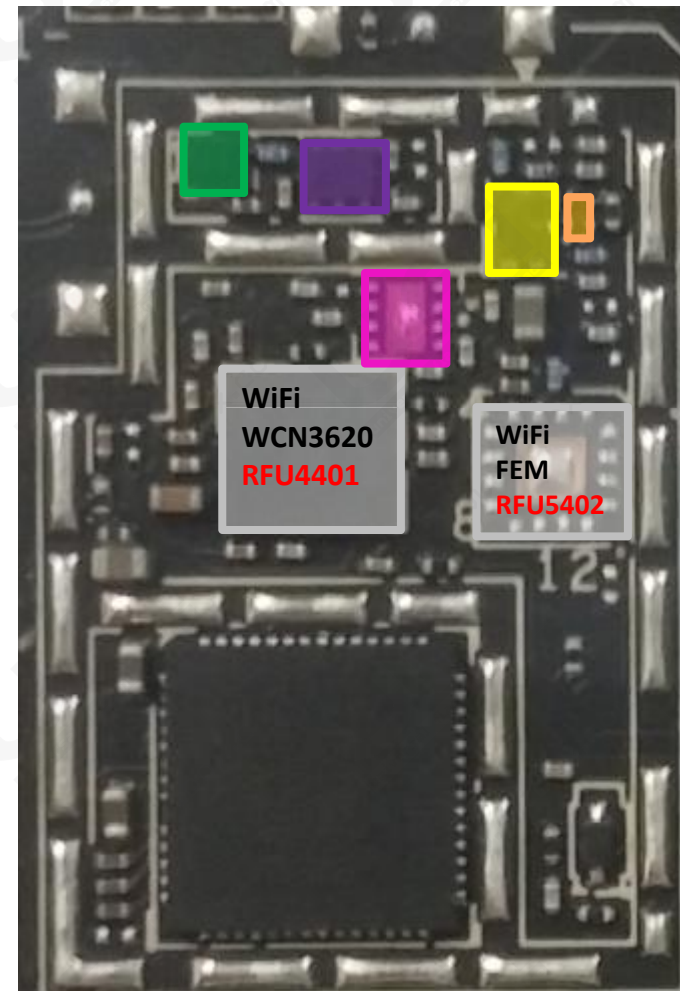


Spring (Phone Main ant.)

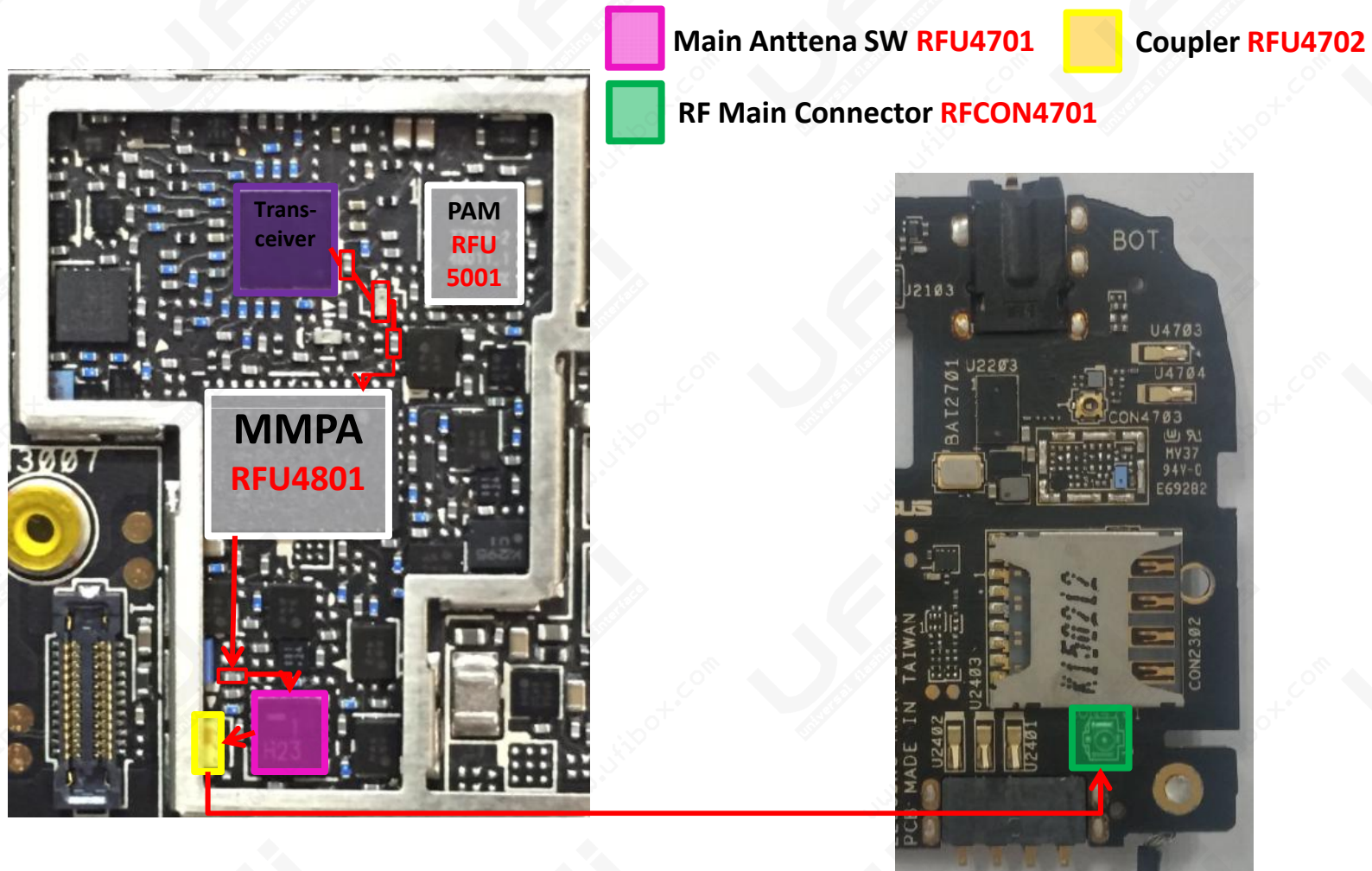
Key Components : WiFi/BT/GPS



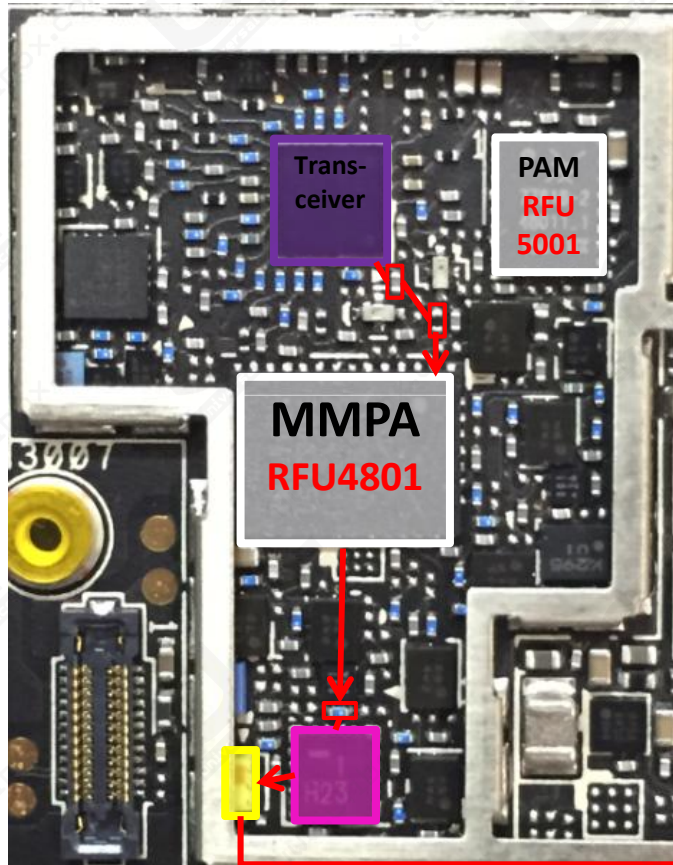
- WiFi/BT Filter **RFU5403**
- WiFi/BT Coupler **RFU5404**
- WiFi/BT Switch **RFU5410**
- GPS 1st Filter **RFU5503**
- GPS LNA **RFU5504**



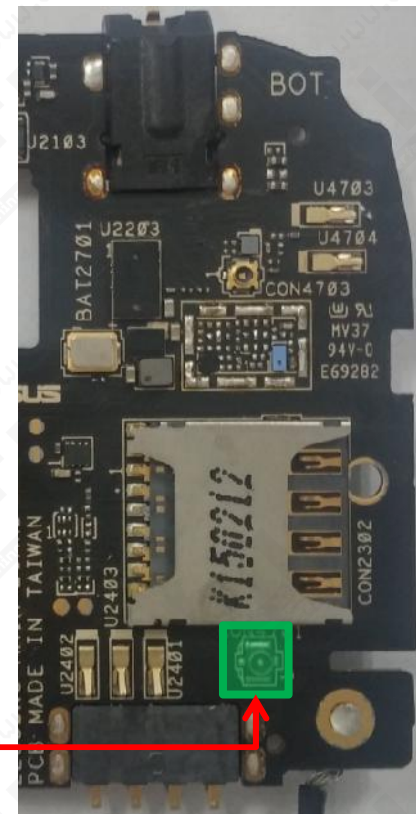
TX Path : 2G GSM 850/GSM 900



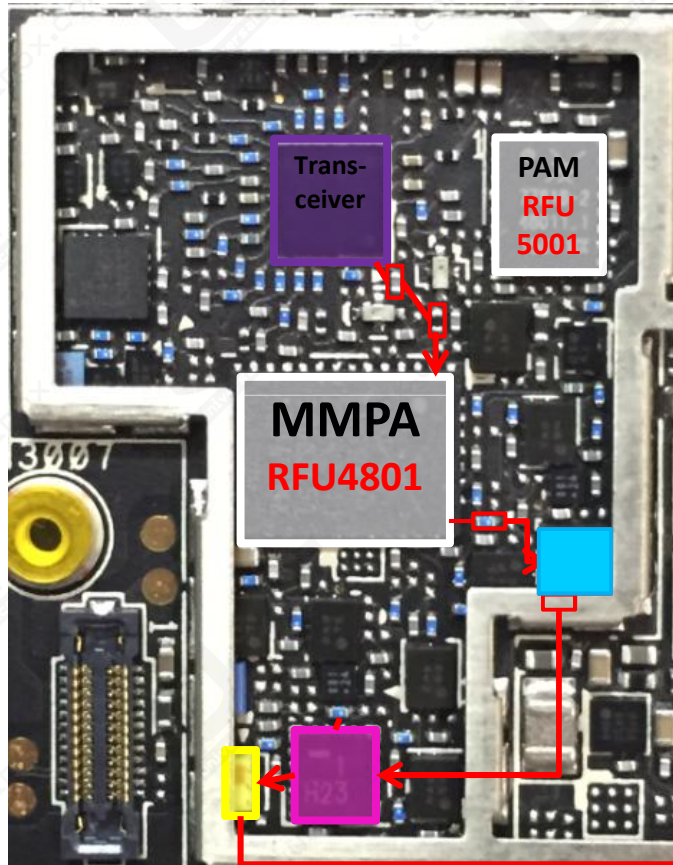
TX Path : 2G DCS/PCS



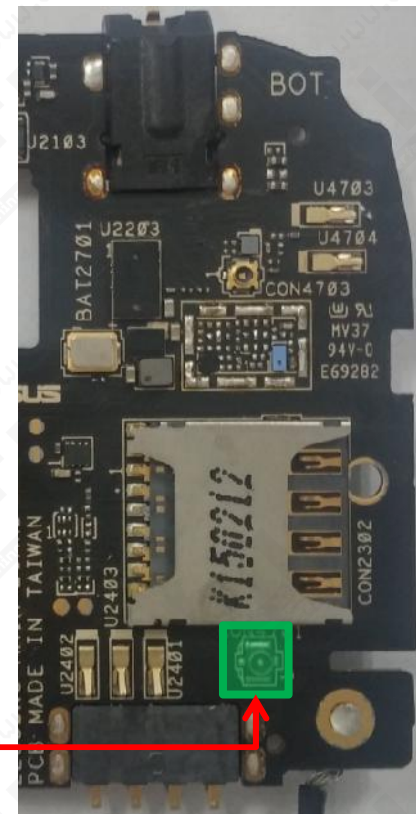
- Main Antenna SW RFU4701** (pink box)
- Coupler RFU4702** (yellow box)
- RF Main Connector RCON4701** (green box)



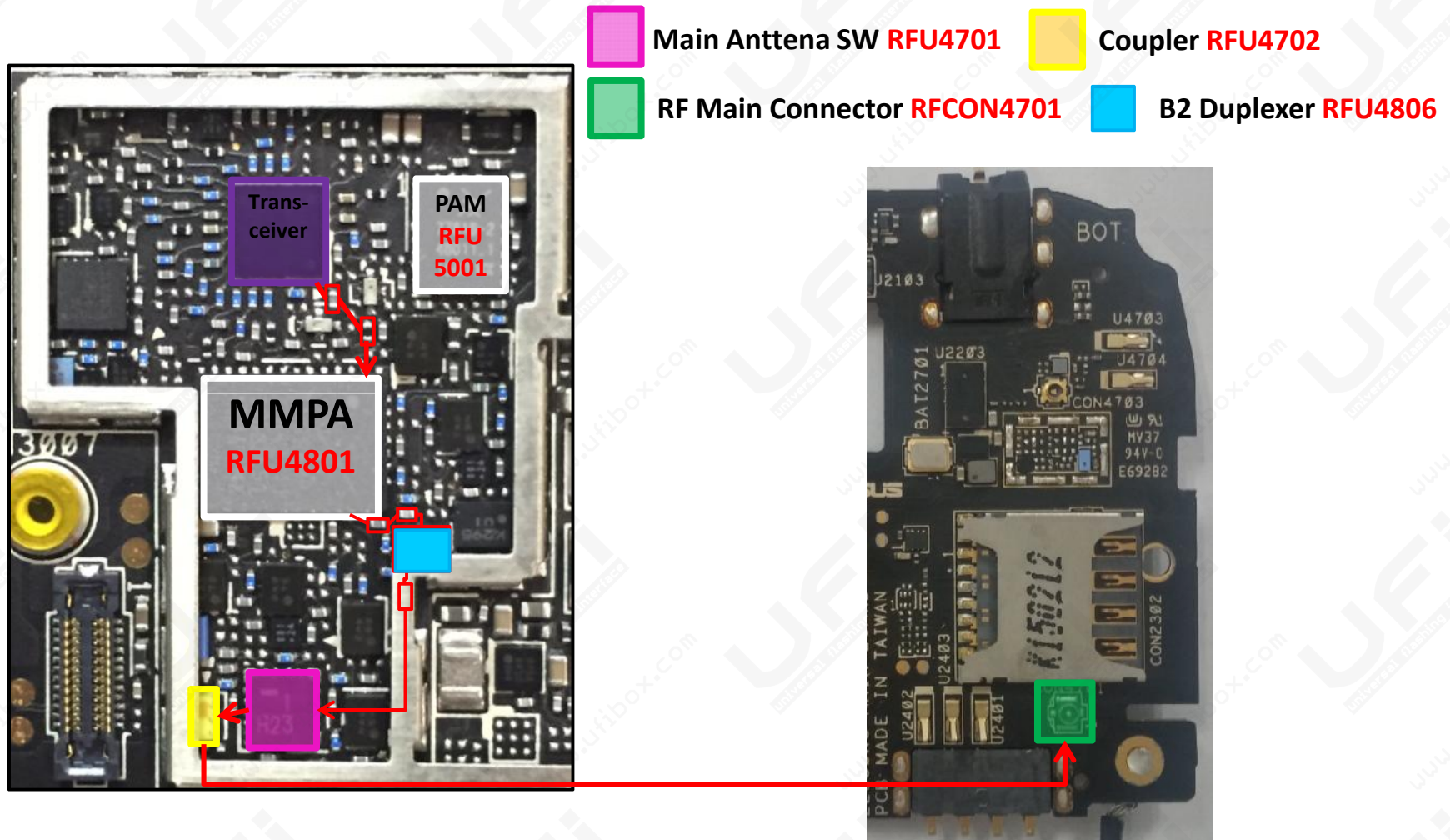
TX Path : 3G&4G Band 1



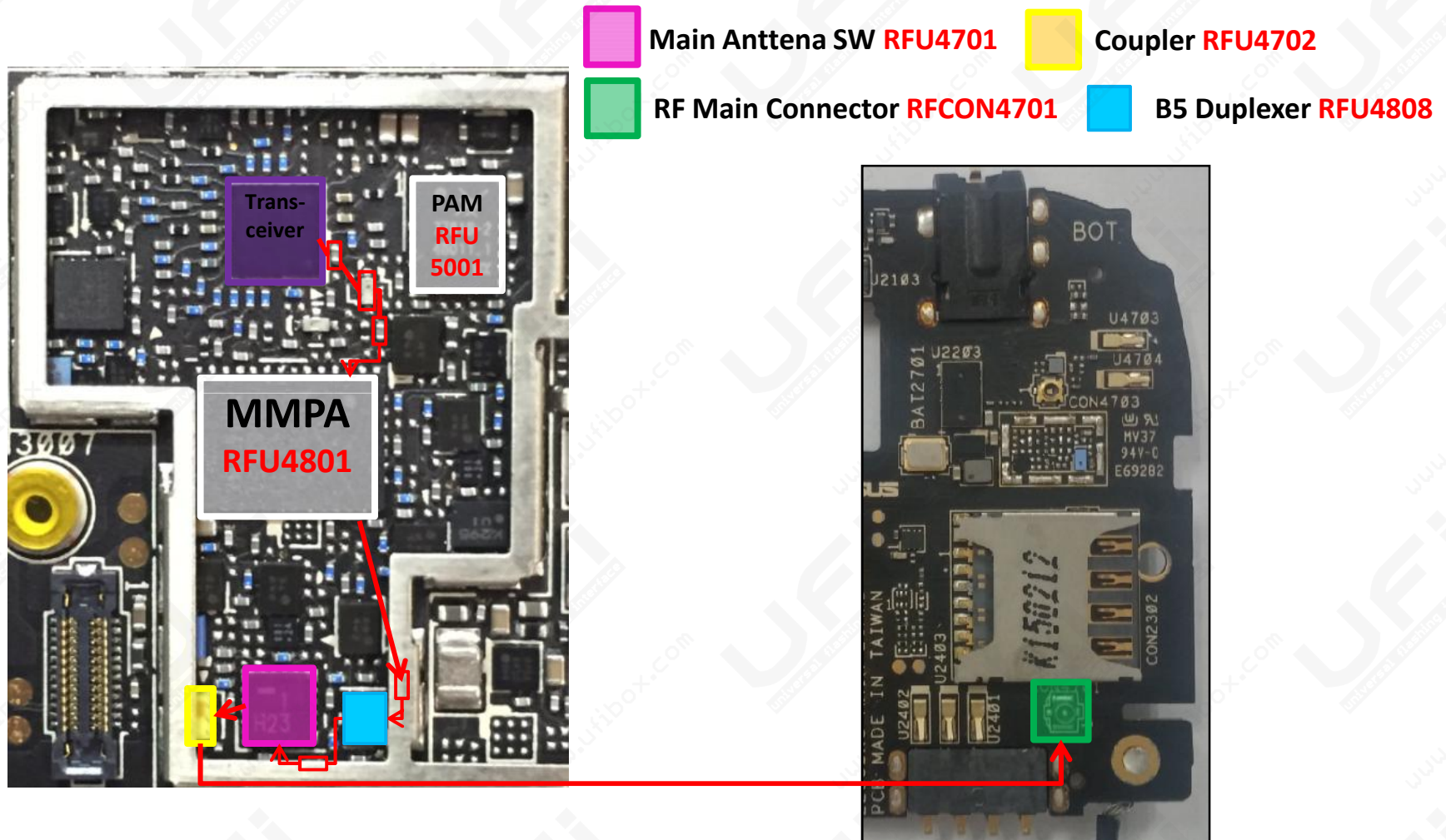
- Main Antenna SW RFU4701
- Coupler RFU4702
- RF Main Connector RFCON4701
- B1 Duplexer RFU4805



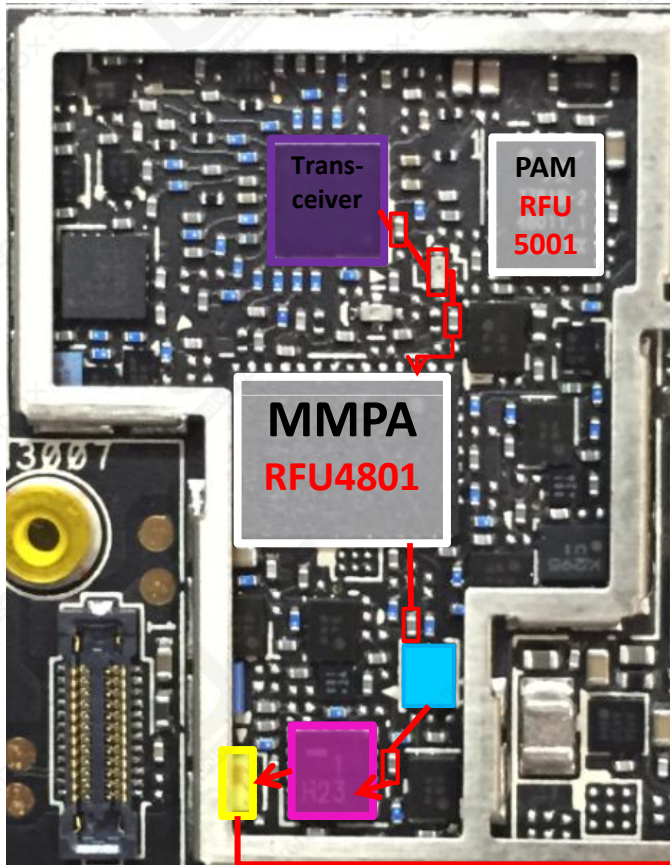
TX Path : 3G&4G Band 2



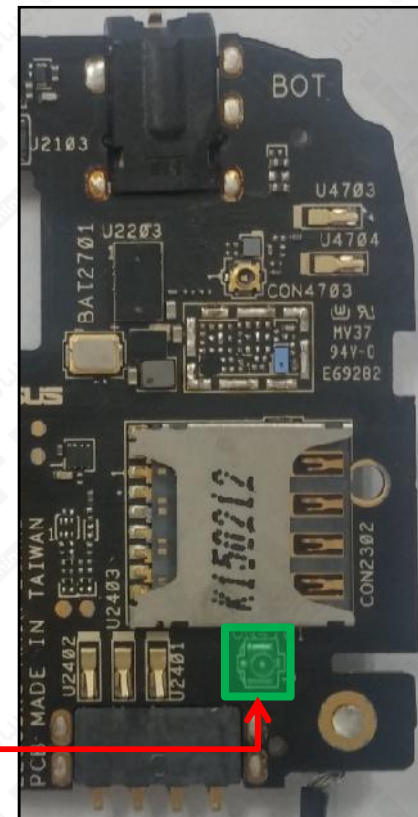
TX Path : 3G&4G Band 5/6/18/19



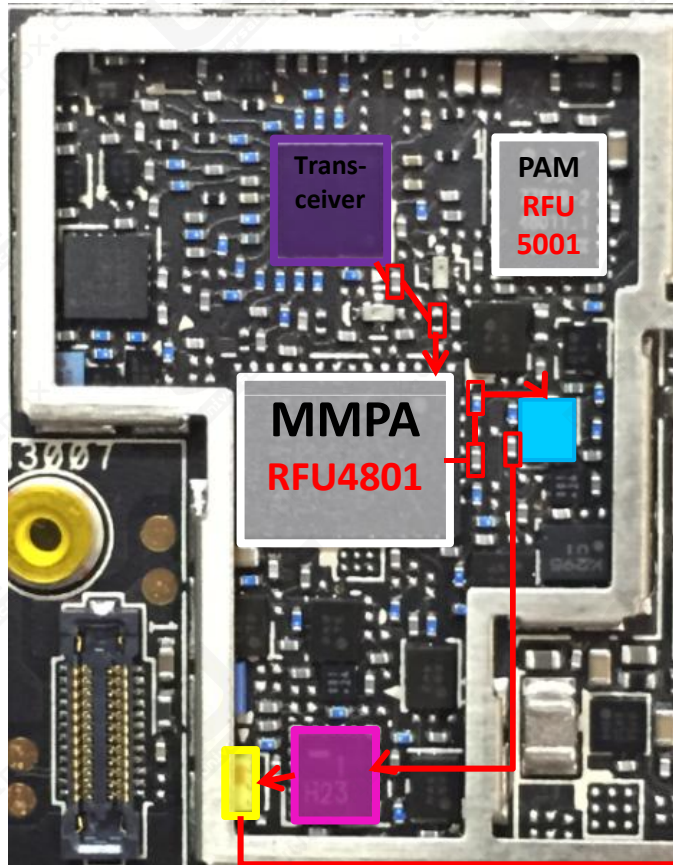
TX Path : 3G&4G Band 8



- Main Antenna SW RFU4701
- Coupler RFU4702
- RF Main Connector RFCON4701
- B8 Duplexer RFU4809



TX Path : 3G&4G Band 3/9



Main Antenna SW RFU4701



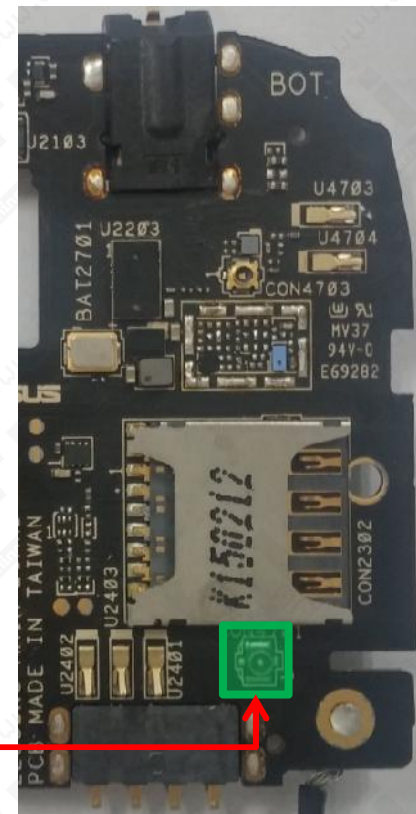
Coupler RFU4702



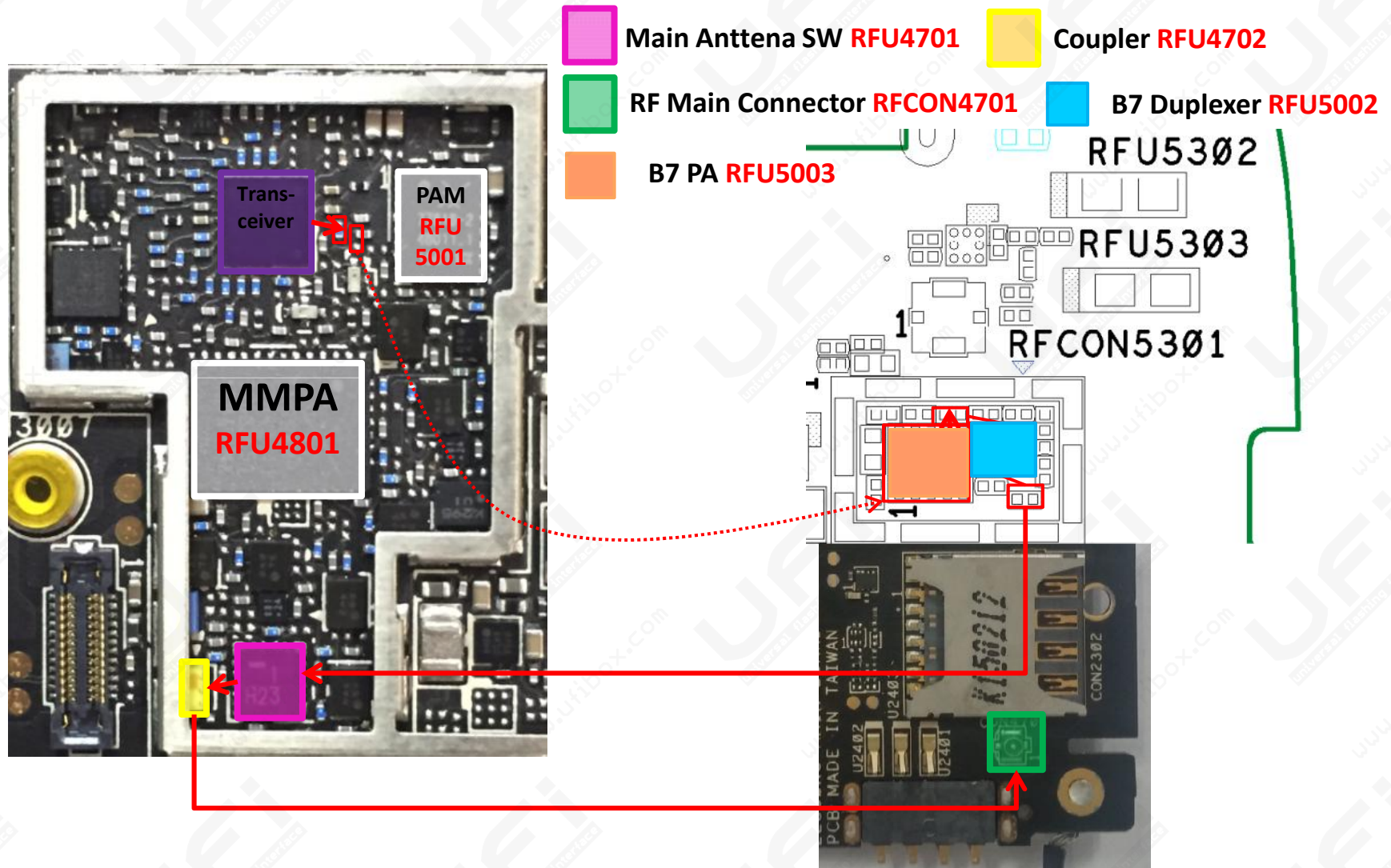
RF Main Connector RFCON4701



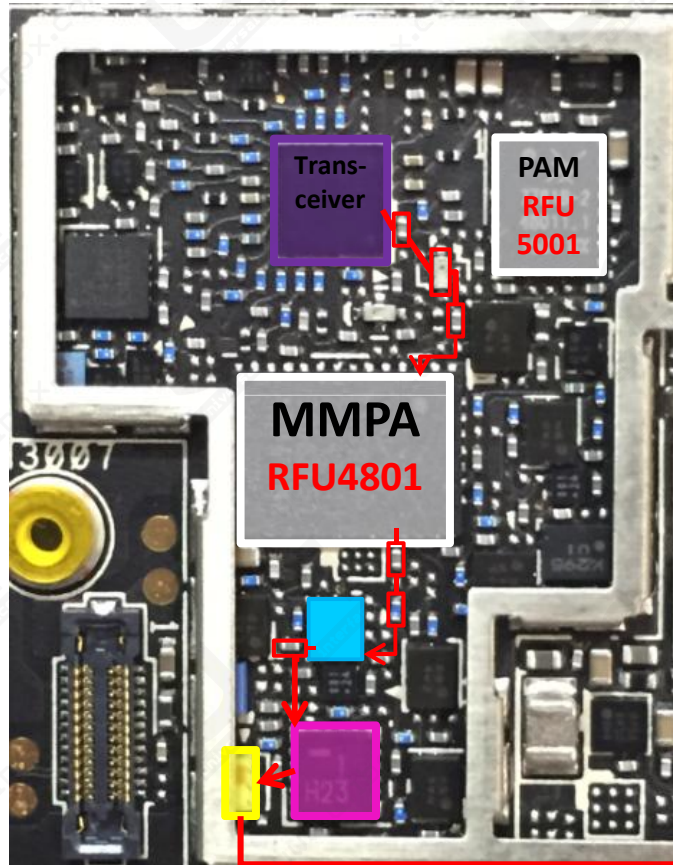
B3 Duplexer RFU4807



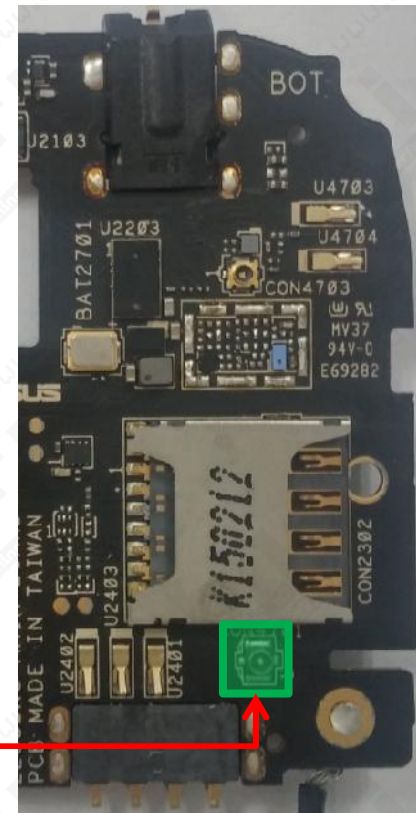
TX Path : 4G Band 7 (WW SKU)



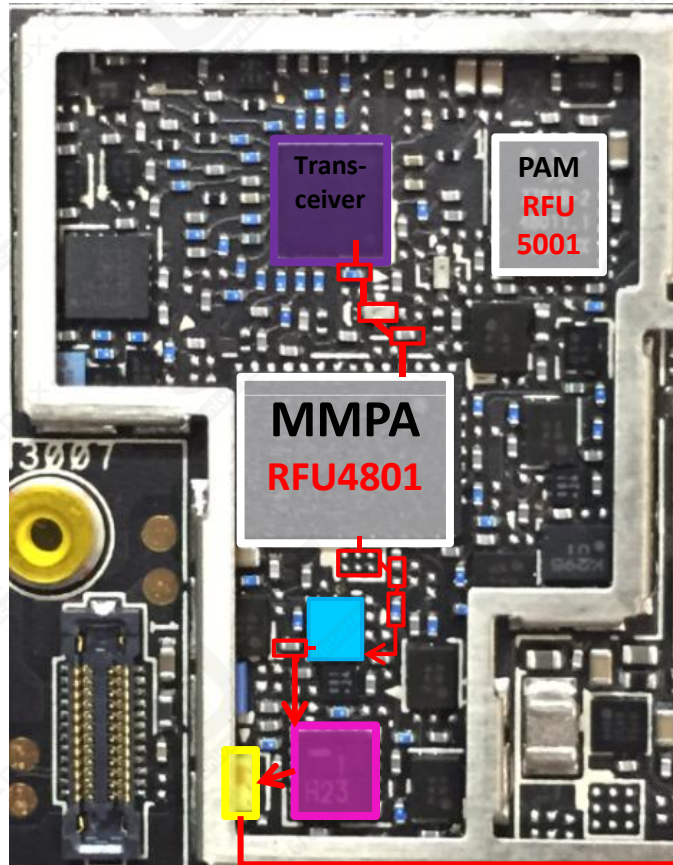
TX Path : 4G Band 20 (WW SKU)



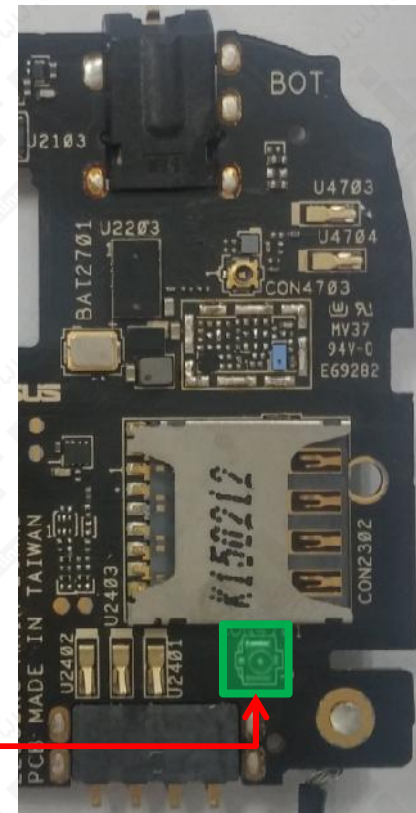
- Main Antenna SW RFU4701
- Coupler RFU4702
- RF Main Connector RFCON4701
- B28B/B20 Duplexer RFU4811



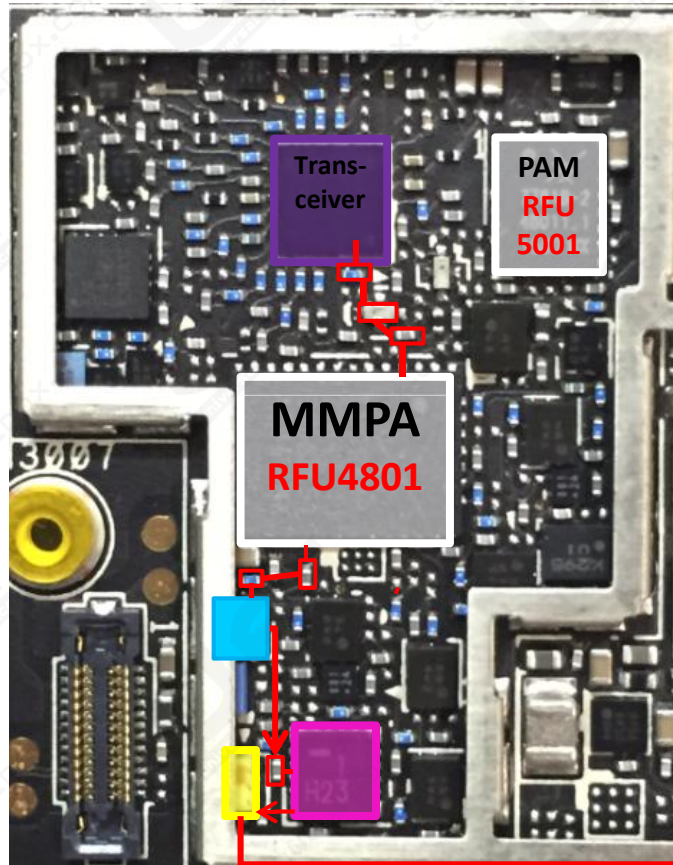
TX Path : 4G Band 28B (TW SKU)



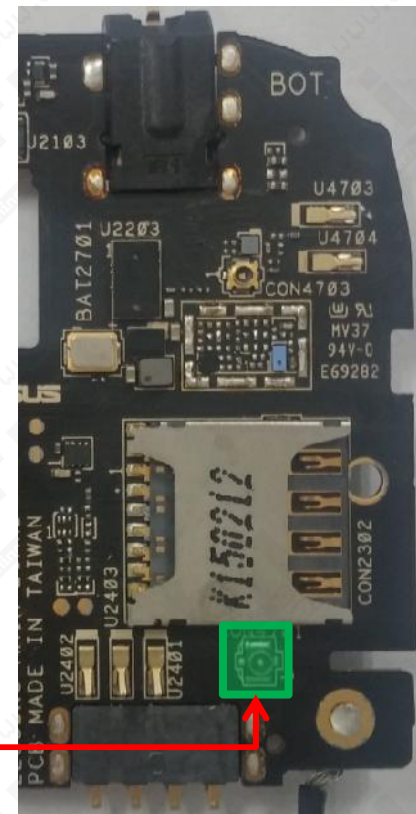
- Main Antenna SW RFU4701
- Coupler RFU4702
- RF Main Connector RFCON4701
- B28B/B20 Duplexer RFU4811



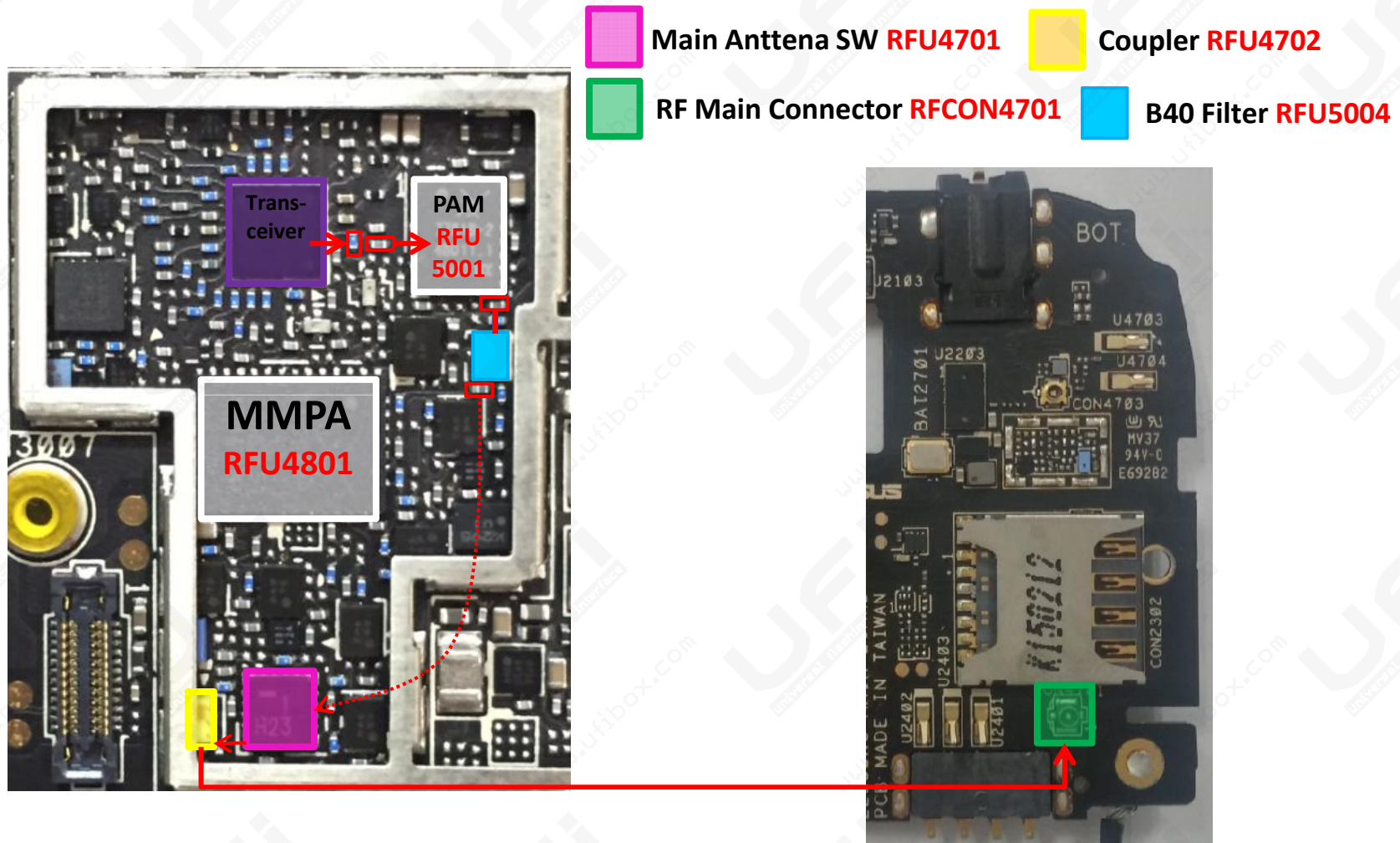
TX Path : 4G Band 28A (TW SKU)



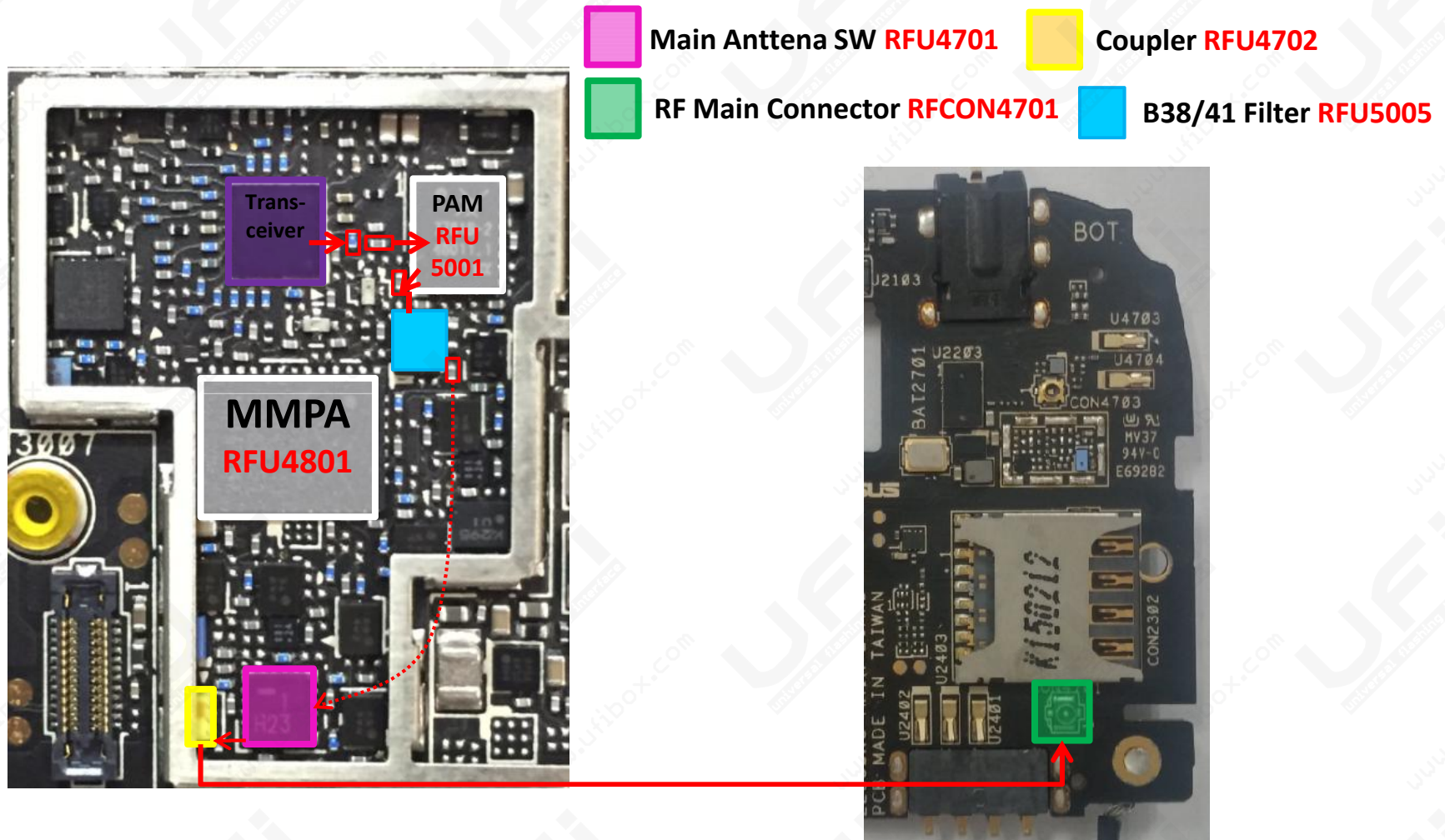
- Main Antenna SW **RFU4701**
- Coupler **RFU4702**
- RF Main Connector **RFCON4701**
- B28A Duplexer **RFU4810**



TX Path : 4G Band 40 (CUCC SKU)

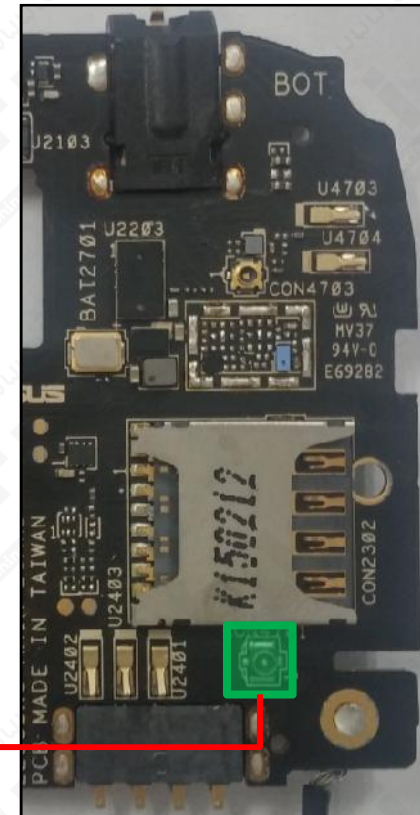


TX Path : 4G Band 38/41 (CUCC SKU)

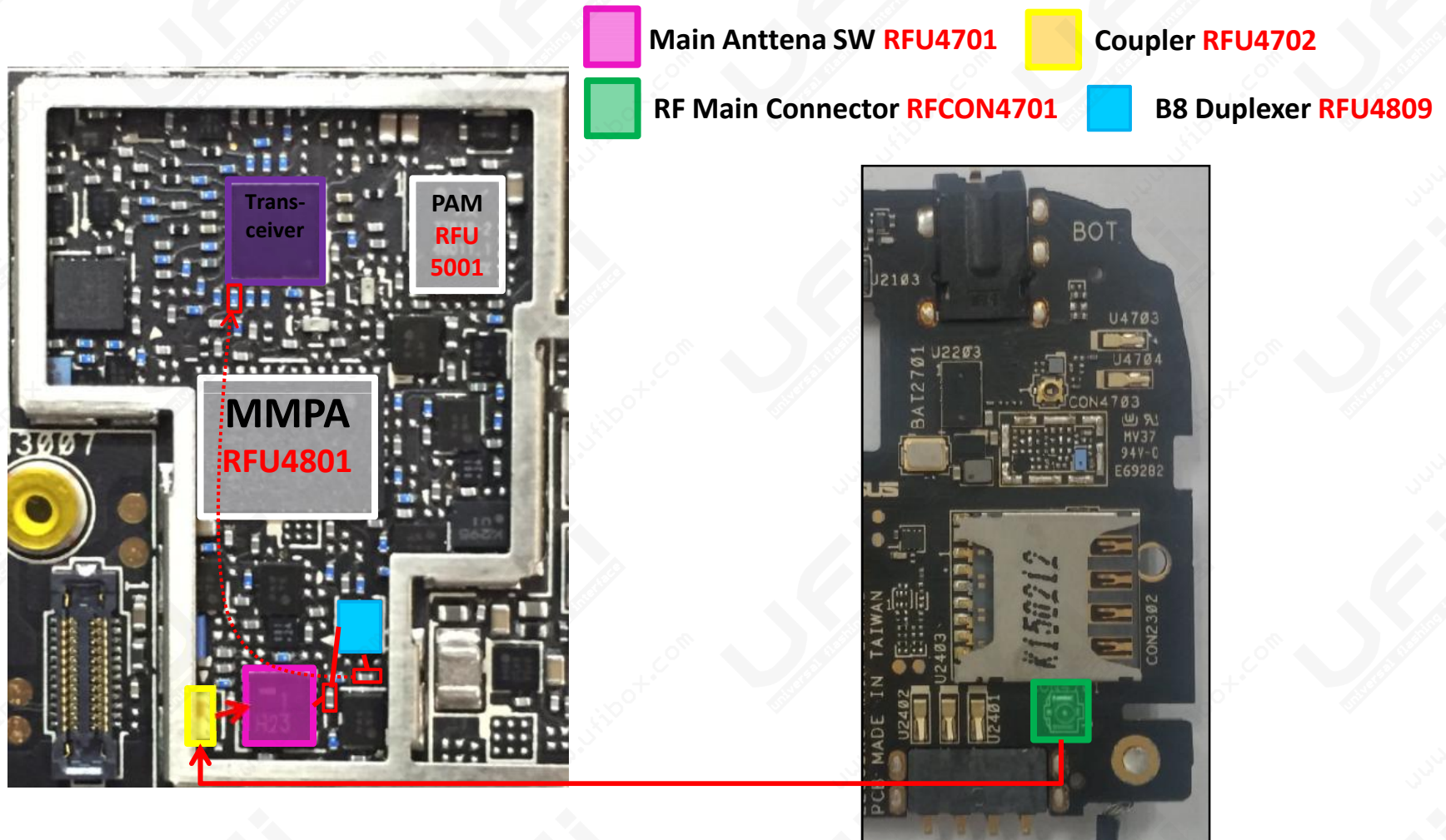


This micrograph shows the MMPPA RFU4801 chip. A red dashed line traces a path from the bottom left towards the center. A yellow box highlights a component at the bottom left, with a red arrow pointing to a pink box labeled 'MMPPA RFU4801'. A purple box labeled 'Transceiver' is located above the pink box. A white box labeled 'PAM RFU 5001' is located to the right of the pink box. A red solid line with arrows points from the bottom edge towards the pink box.

B5 Duplexer RFU4808



RX Path : 2G GSM 900/3G Band 8/4G Band 8



Transceiver

PAM
RFU
5001

MMPA
RFU4801

7

423

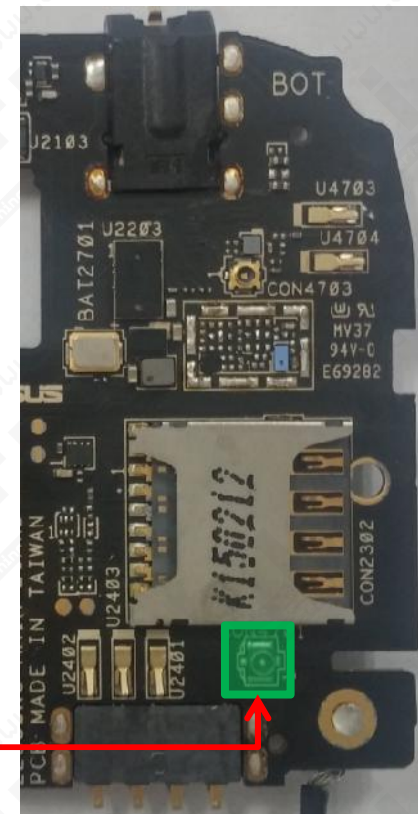
Main Antenna SW RFU4701

Coupler RFU4702

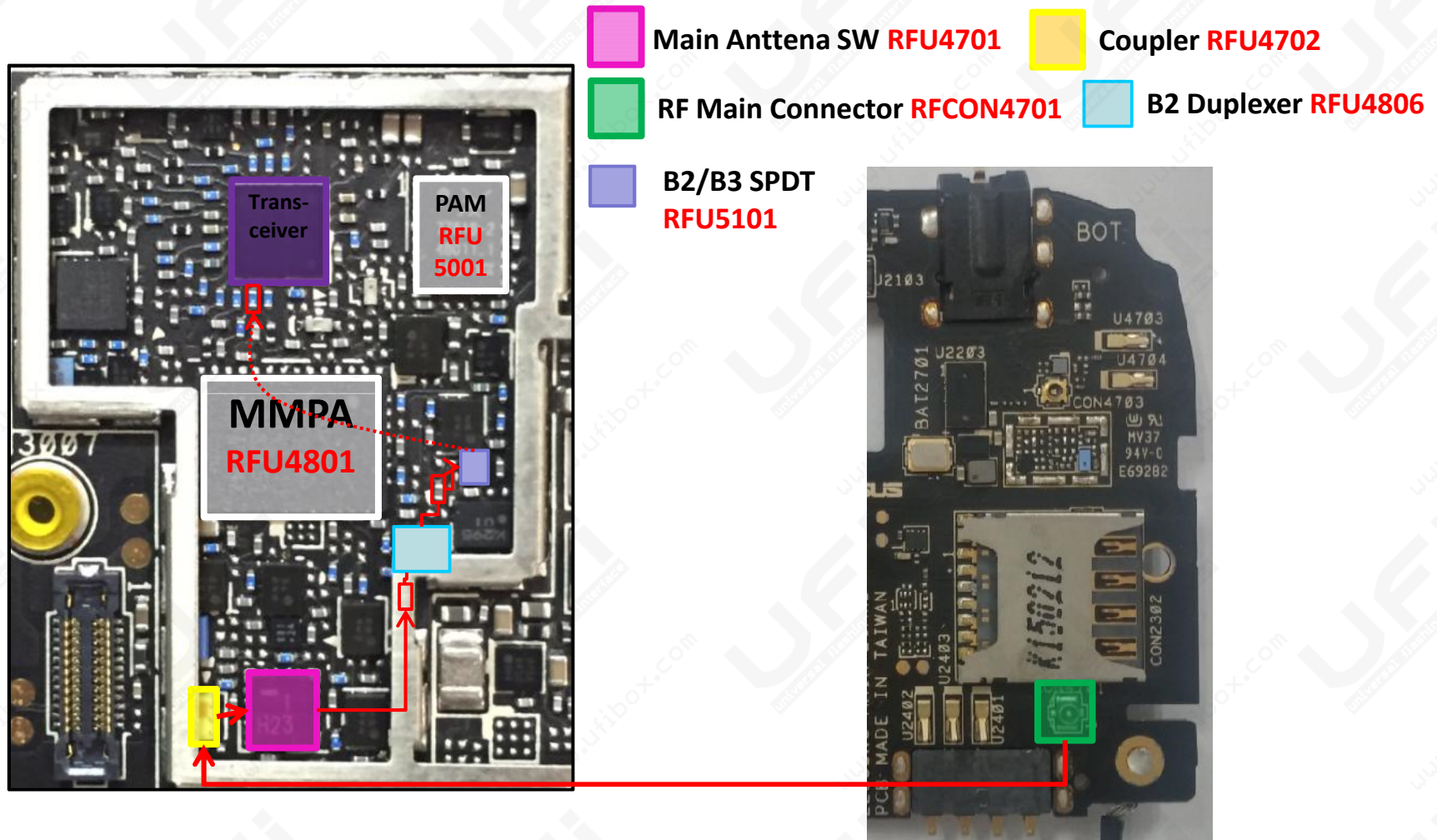
RF Main Connector RCON4701

B3 Duplexer RFU4807

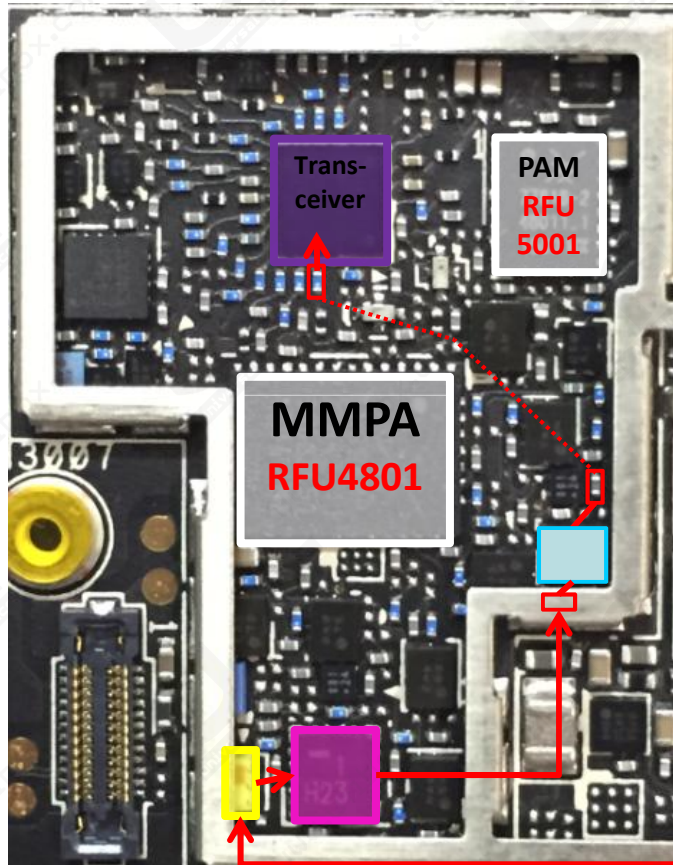
B2/B3 SPDT
RFU5101



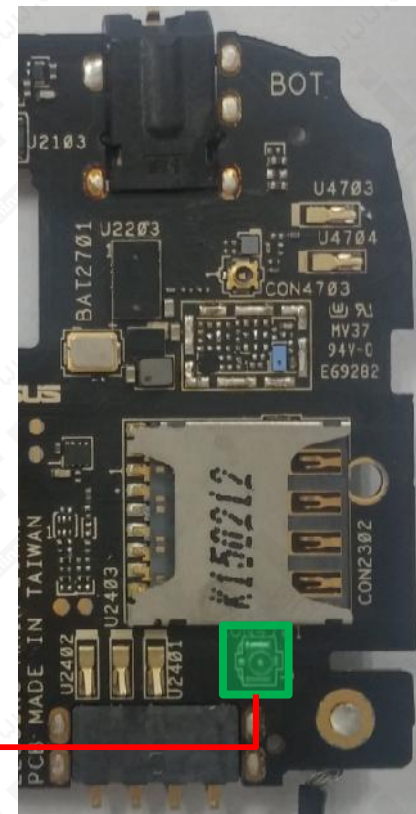
RX Path : 2G PCS/3G Band 2



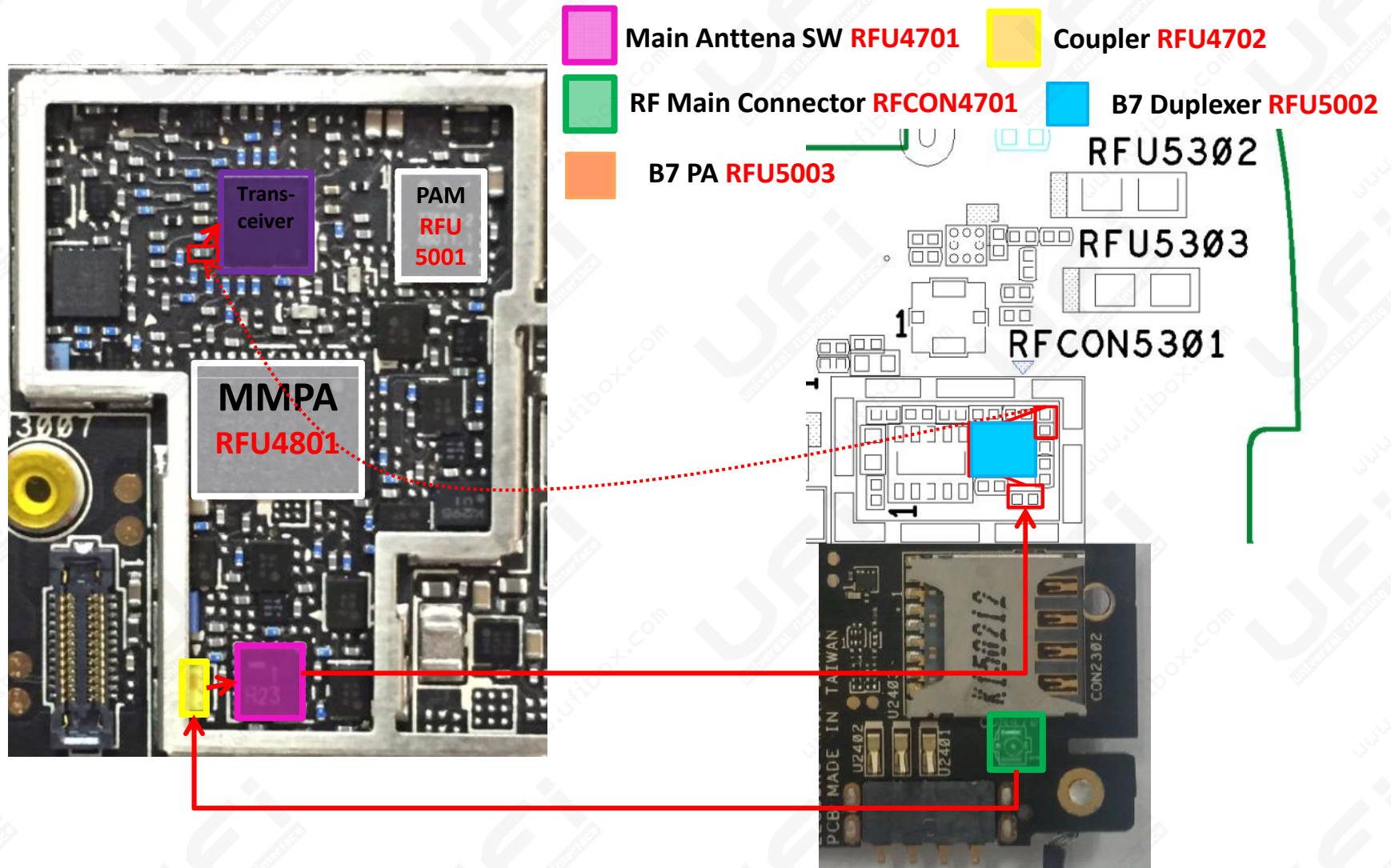
RX Path : 3G&4G Band 1



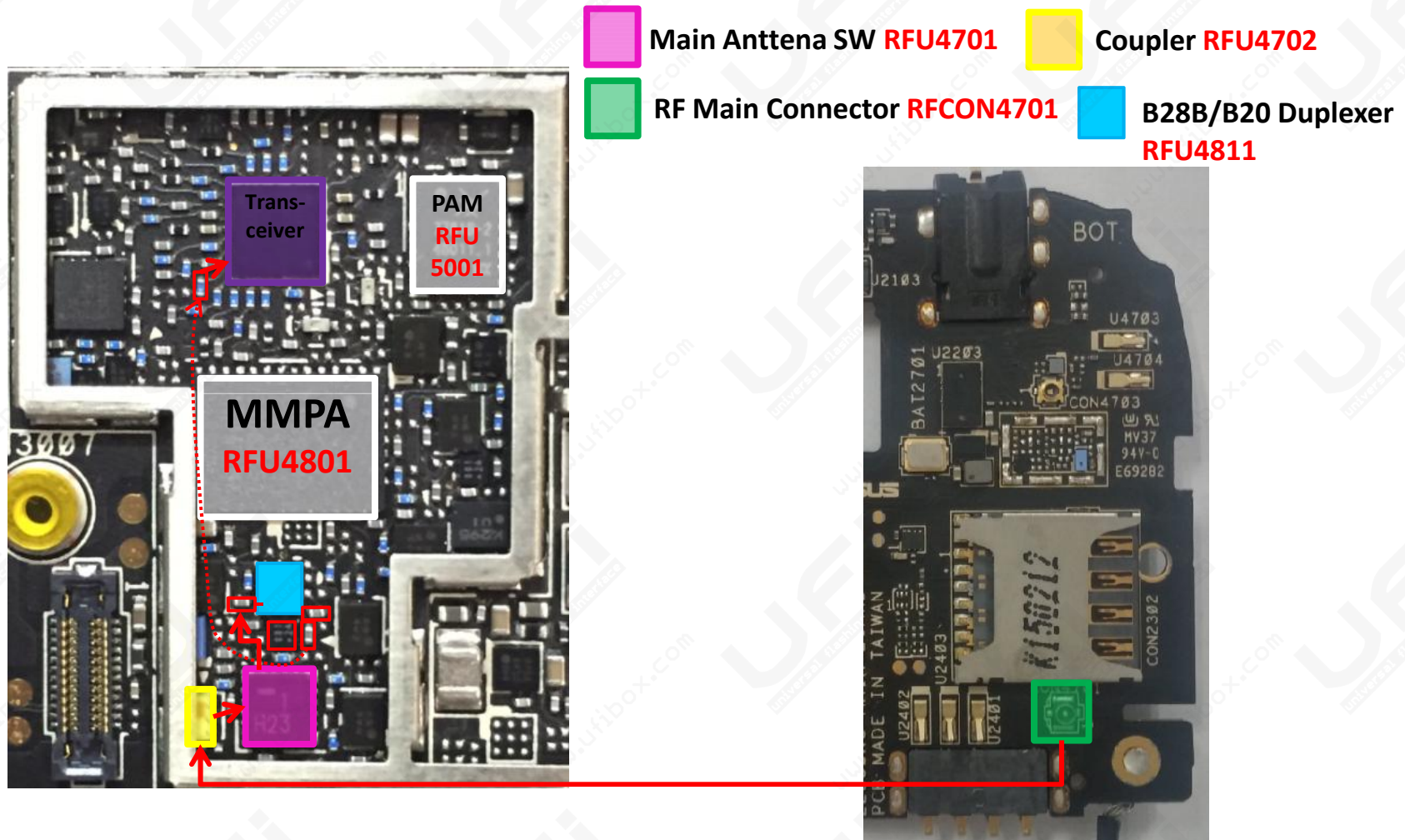
- Main Antenna SW RFU4701
- Coupler RFU4702
- RF Main Connector RFCON4701
- B1 Duplexer RFU4805



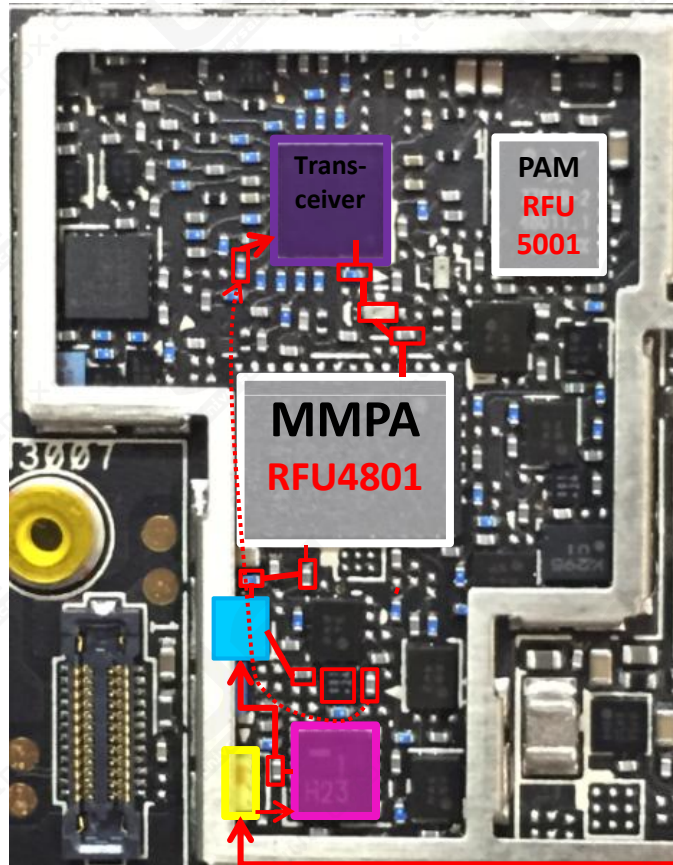
RX Path : 4G Band 7 (WW SKU)



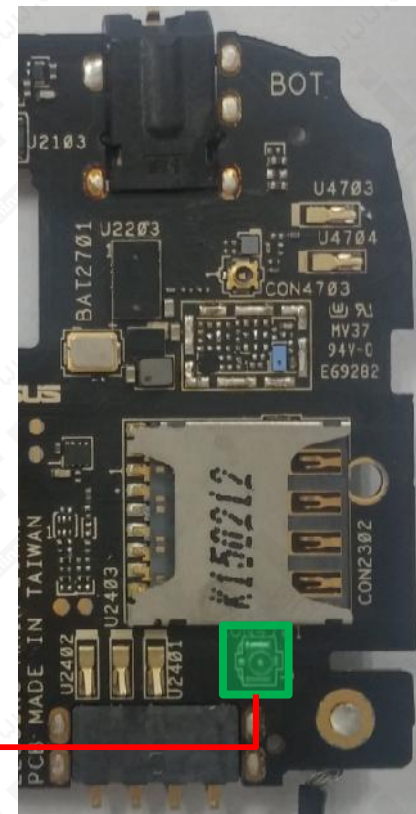
RX Path : 4G Band 20/28B (WW SKU)



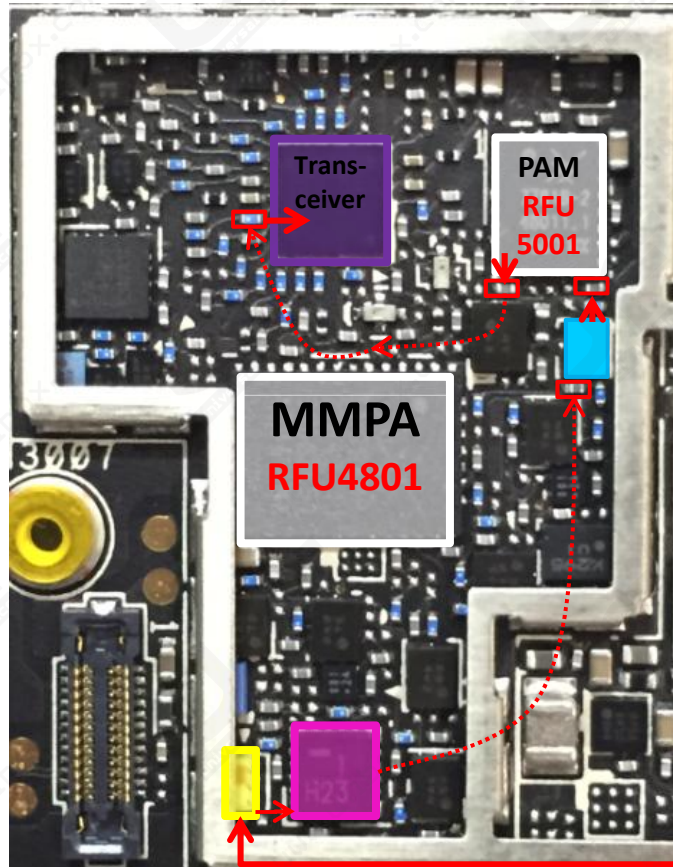
RX Path : 4G Band 28A (TW SKU)



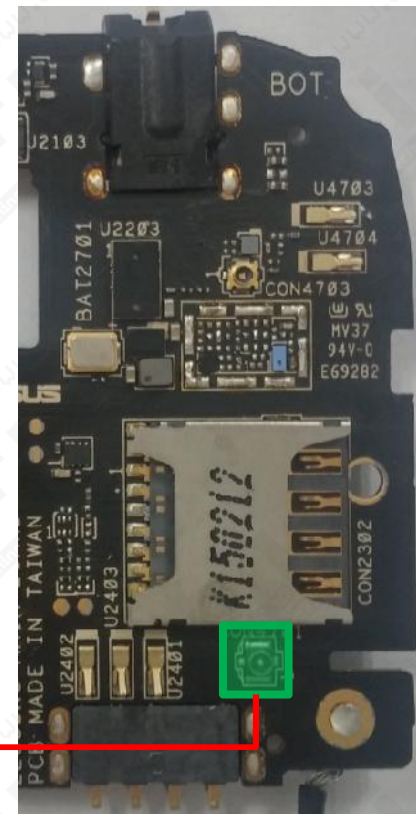
- Main Antenna SW RFU4701
- Coupler RFU4702
- RF Main Connector RFCON4701
- B28A Duplexer RFU4810



RX Path : 4G Band 40 (CUCC SKU)



- Main Antenna SW RFU4701
- Coupler RFU4702
- RF Main Connector RFCON4701
- B40 Filter RFU5004



A photograph of a mobile phone circuit board with several components highlighted by colored boxes and labeled. A red dotted line traces a signal path through the board. The components and their labels are:

- Transceiver**: A purple box labeled "Transceiver" with a red arrow pointing to it from the left.
- PAM RFU 5001**: A grey box labeled "PAM RFU 5001" with a red arrow pointing to it from the left.
- MMPA RFU4801**: A grey box labeled "MMPA RFU4801" with a red arrow pointing to it from the left.
- 423**: A pink box labeled "423" with a red arrow pointing to it from the left.

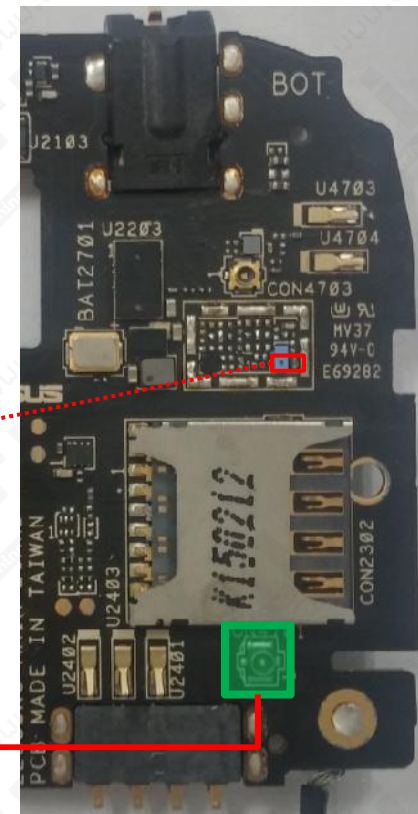
The red dotted line starts at the bottom left, goes up to the "423" component, then continues up to the "MMPA RFU4801" component, then continues up to the "PAM RFU 5001" component, and finally continues up to the "Transceiver" component.

Main Antenna SW RFU4701

Coupler RFU4702

RF Main Connector RCON4701

B38/41 Filter RFU5005



Diversity RX Path : 3G/4G Band 1



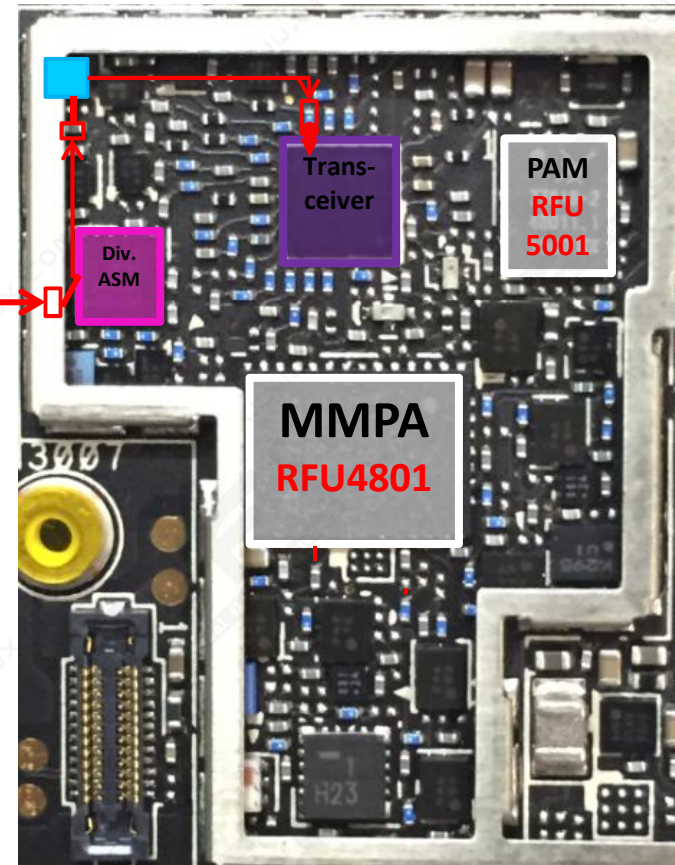
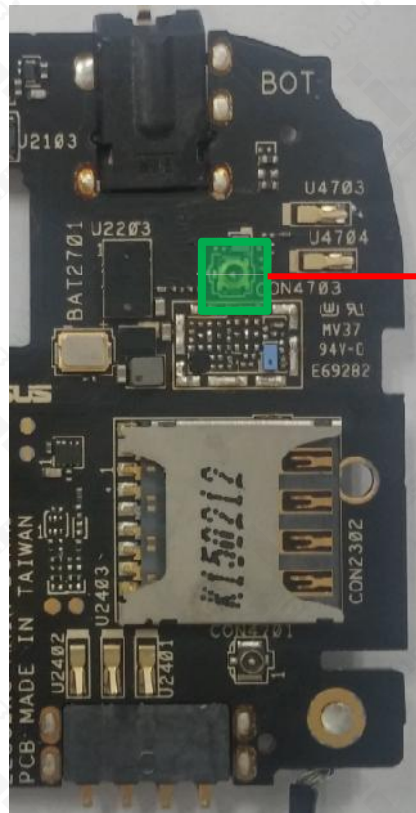
Div. Antenna SW **RFU5301**



B1 Div. Filter **RFU5202**



RF **Diversity** Connector **RFCON5301**



Diversity RX Path : 3G/4G Band 2



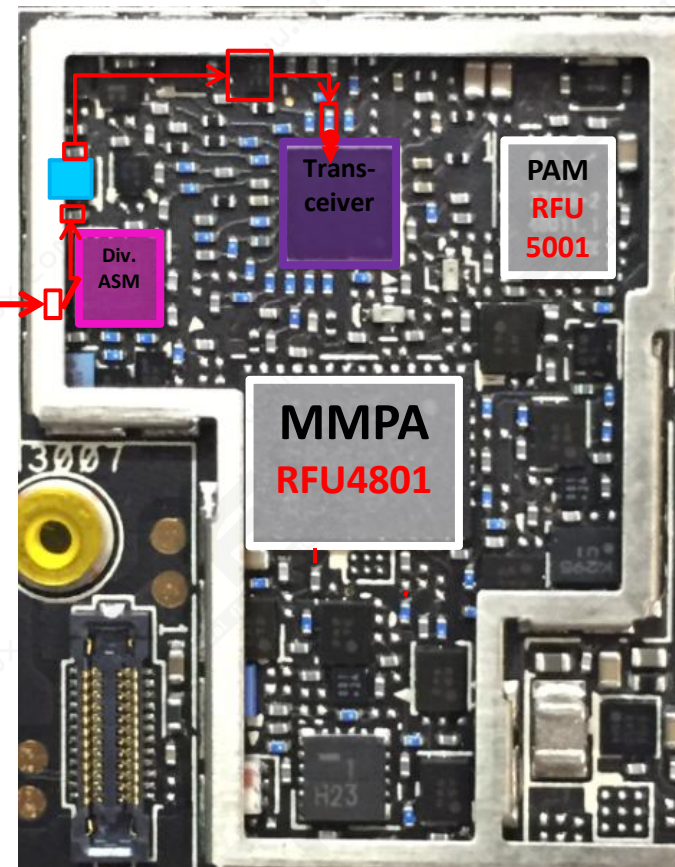
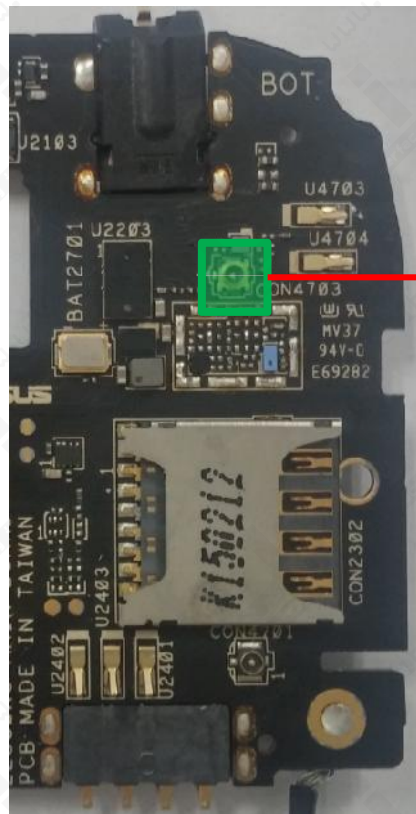
Div. Antenna SW **RFU5301**



B2 Div. Filter **RFU5201**



RF *Diversity* Connector **RFCON5301**



Diversity RX Path : 3G/4G Band 5



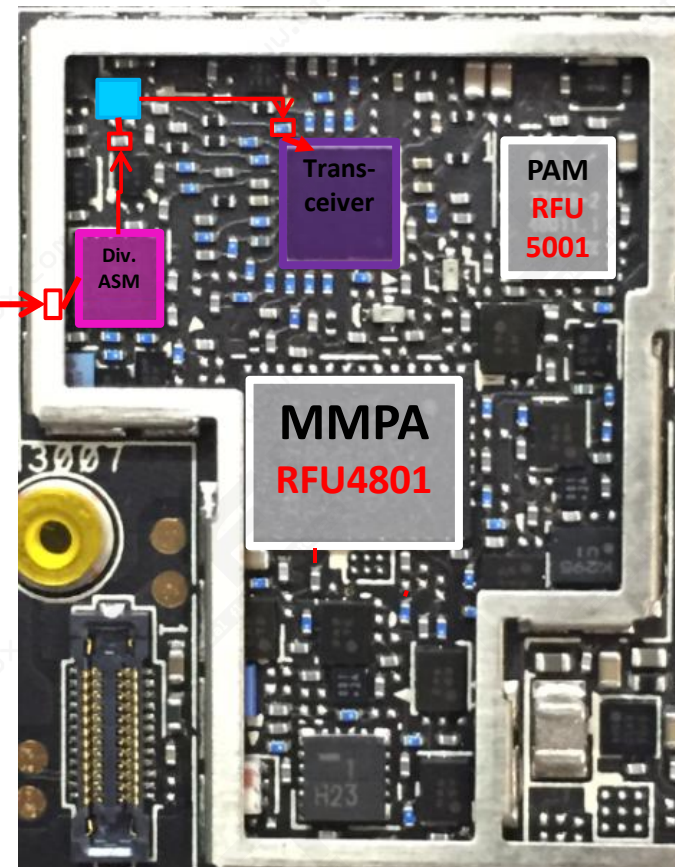
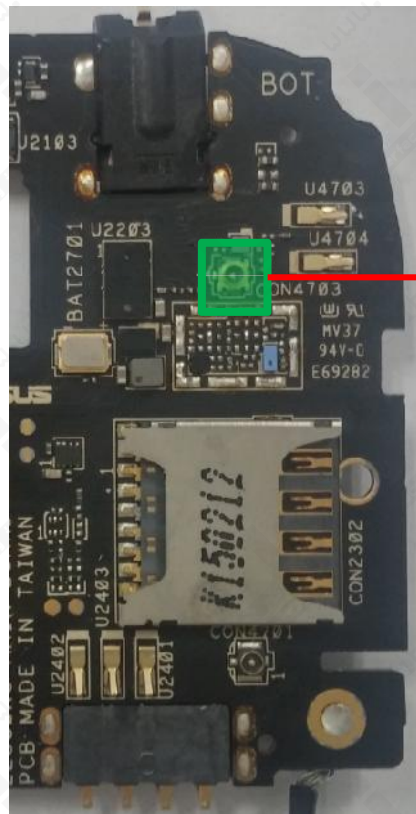
Div. Antenna SW **RFU5301**



B5 Div. Filter **RFU5206**



RF **Diversity** Connector **RFCON5301**



Diversity RX Path : 3G/4G Band 8



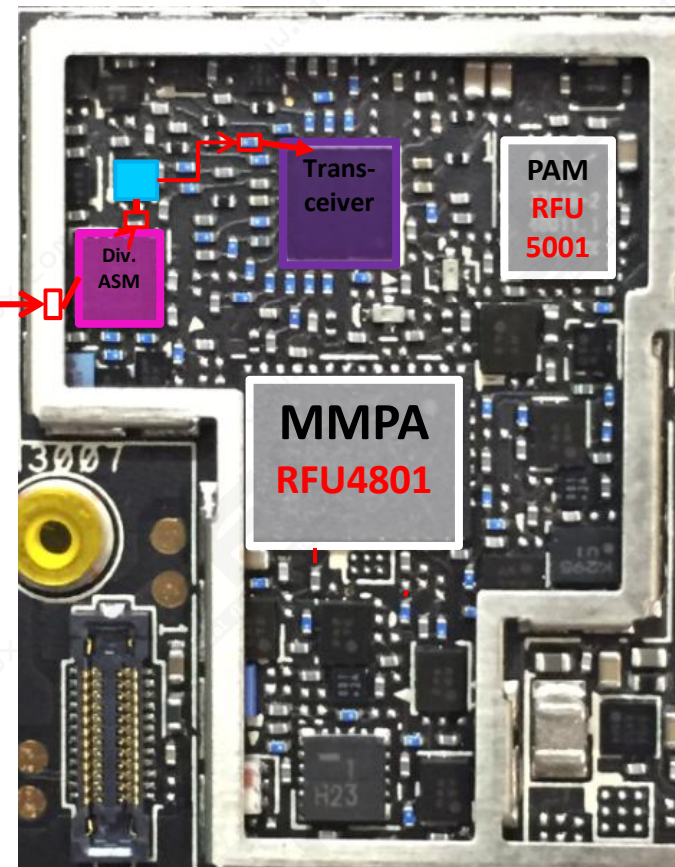
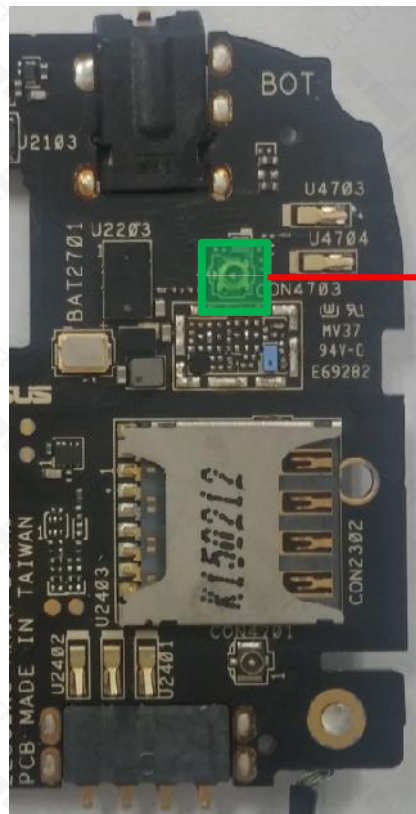
Div. Antenna SW **RFU5301**



B8 Div. Filter **RFU5203**



RF **Diversity** Connector **RFCON5301**



Diversity RX Path : 4G Band 3



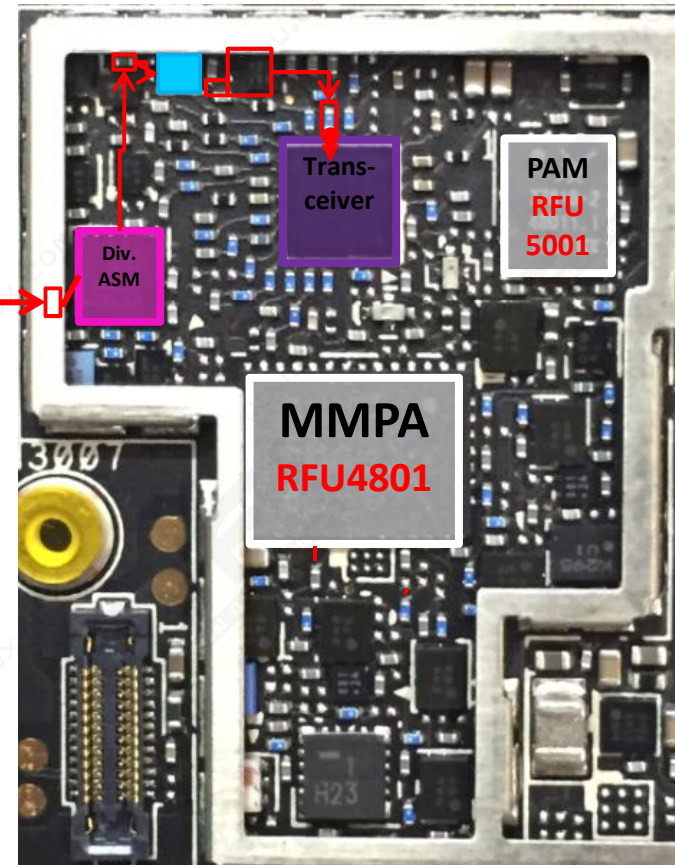
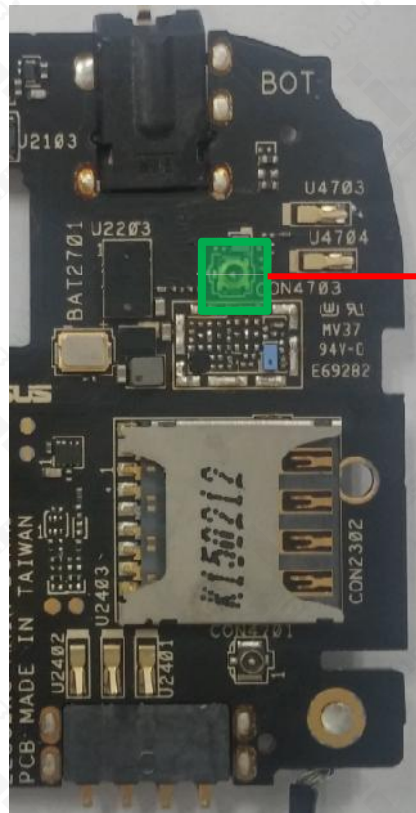
Div. Antenna SW **RFU5301**



B3 Div. Filter **RFU5204**



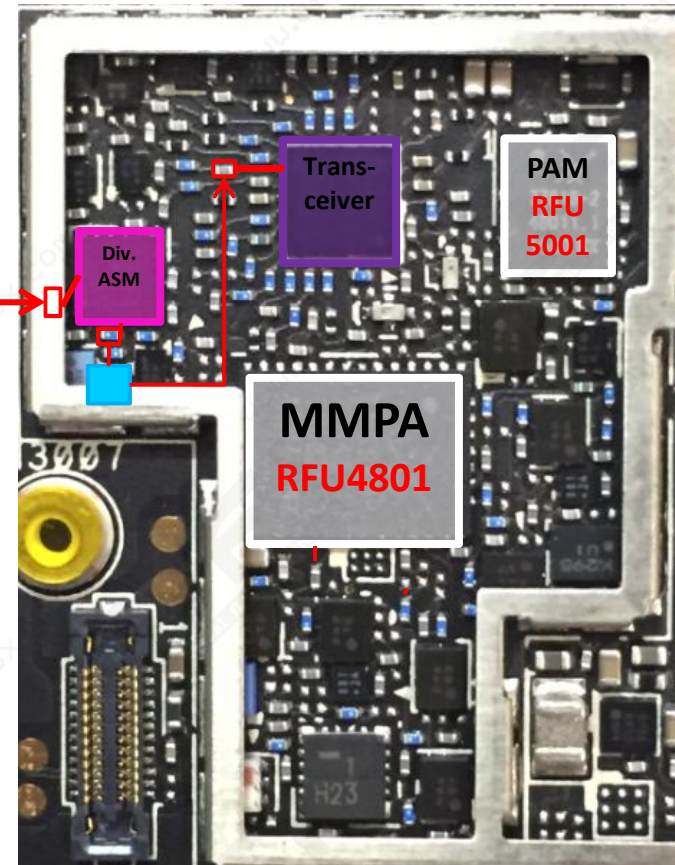
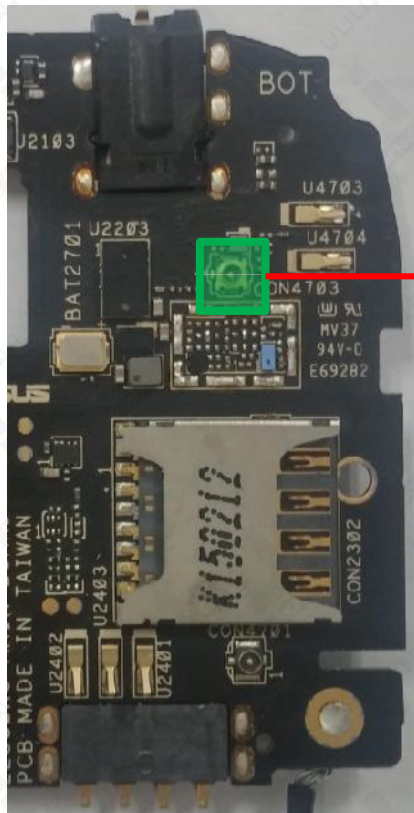
RF **Diversity** Connector **RFCON5301**



Diversity RX Path : 4G Band 7

Div. Antenna SW RFU5301

B7 Div. Filter RFU5209

 RF *Diversity* Connector **RFCON5301**

Diversity RX Path : 4G Band 20 (WW)/ Band 28(TW)



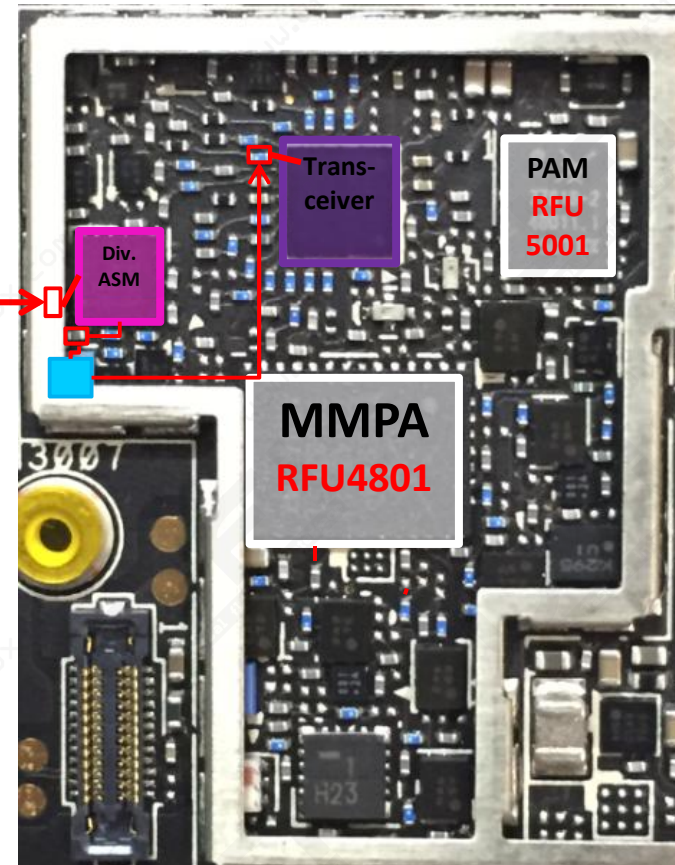
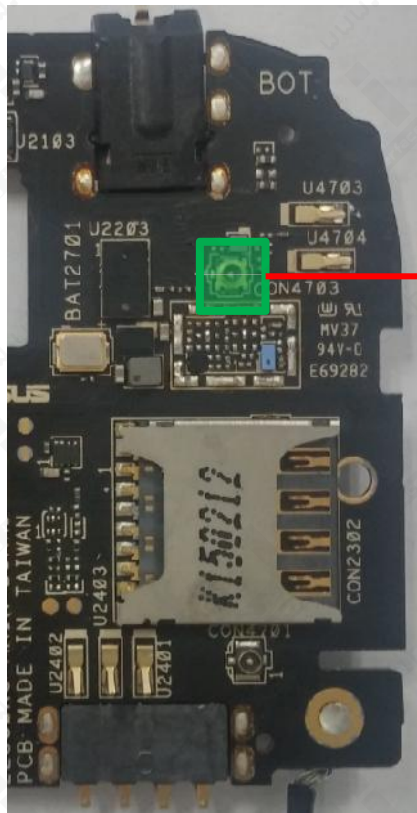
Div. Antenna SW **RFU5301**



B20 Div. Filter **RFU5211**

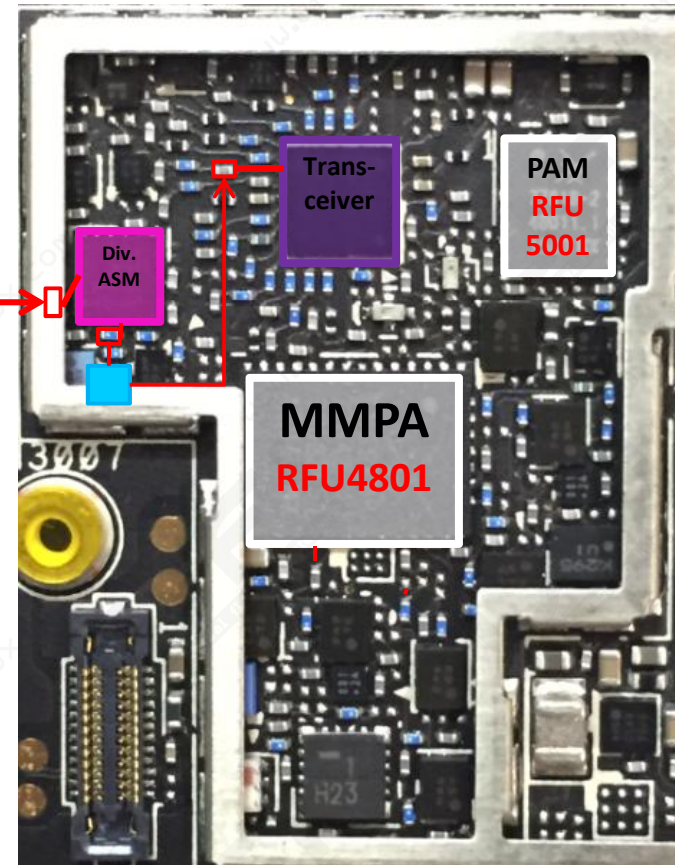
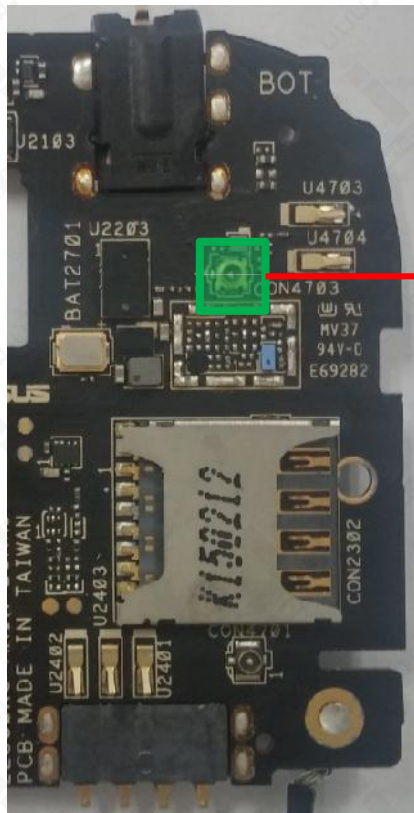


RF **Diversity** Connector **RFCON5301**



Diversity RX Path : 4G Band 38/41

註: B38/41與B7 RF走同路徑但不共存



Diversity RX Path : 4G Band 40



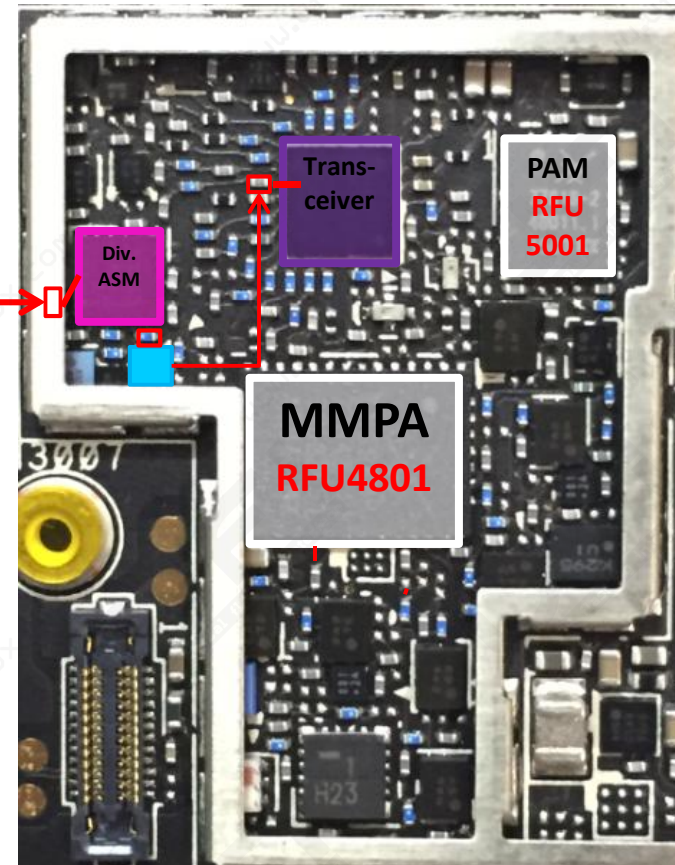
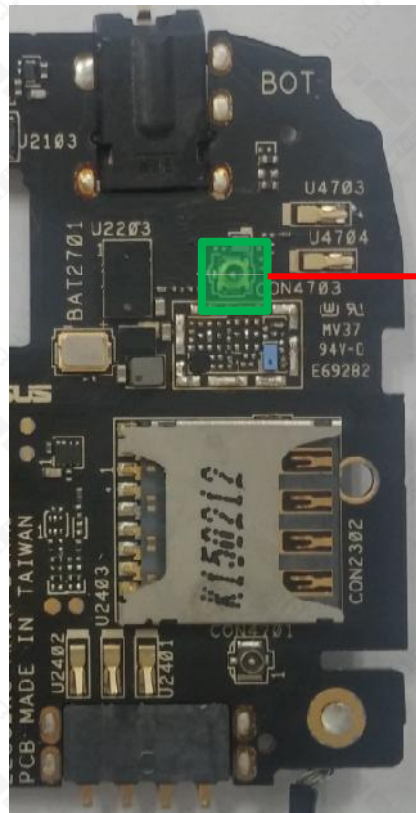
Div. Antenna SW **RFU5301**



B40 Div. Filter **RFU5208**

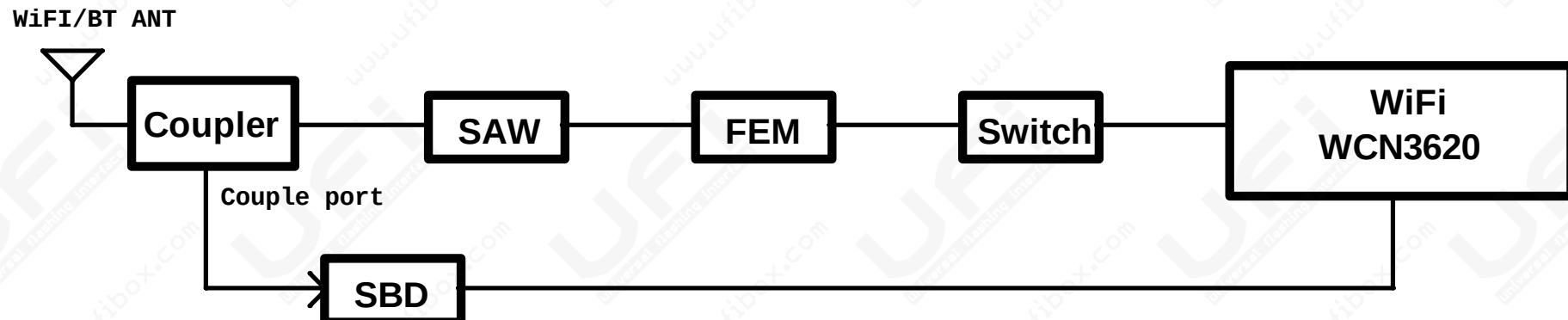


RF **Diversity** Connector **RFCON5301**

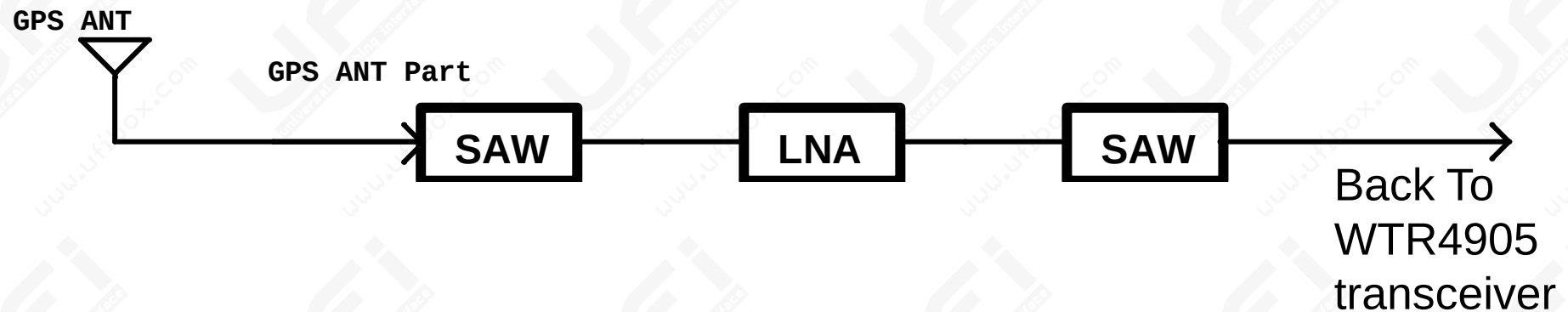


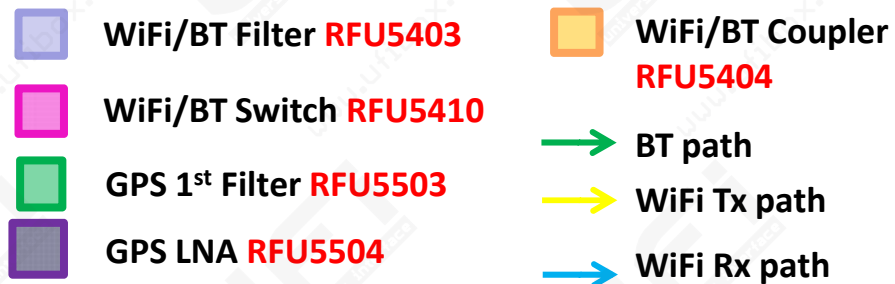
Wi-Fi/BT/GPS block diagram

WiFi/BT part

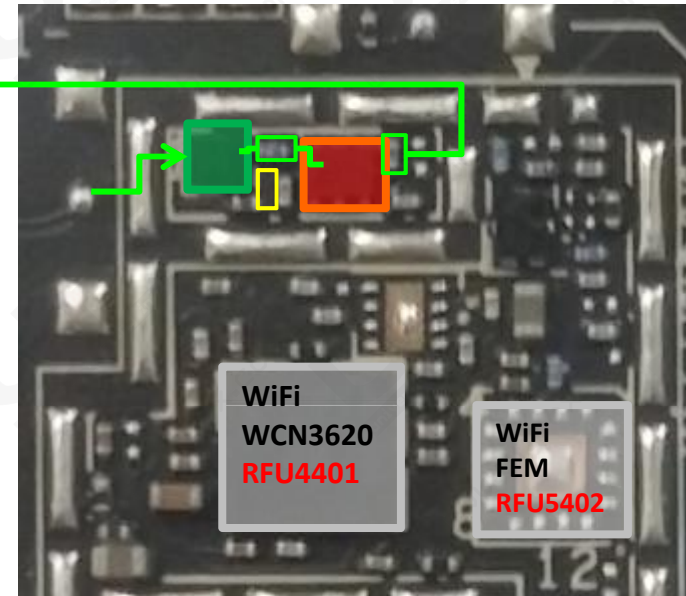
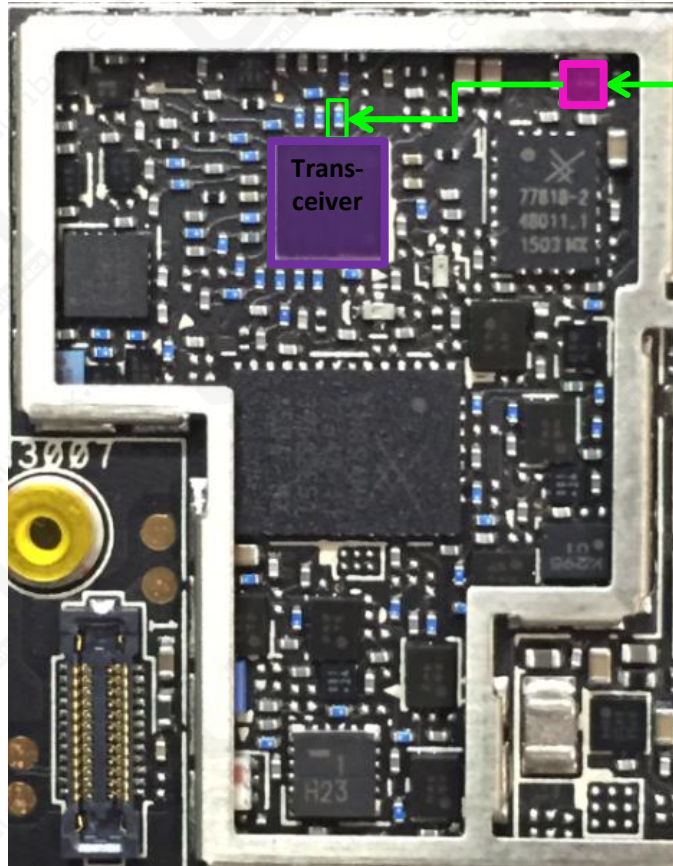


GPS part



[illegible]

Tx/Rx Path: GPS



-  GPS 1st Filter **RFU5503**
-  GPS LNA **RFU5504**
-  GPS 2nd Filter **RFU5506**
-  GPS LNA BEAD **RFL5502** ;
LNA Voltage 1.8V

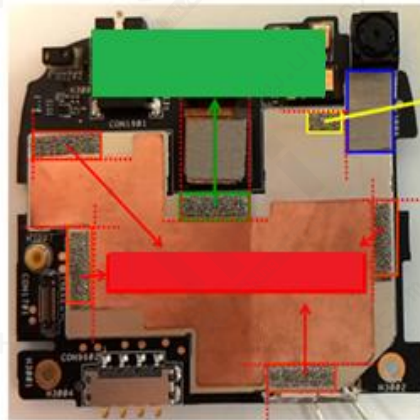
Wi-Fi/BT Function Test Procedure

- Change antenna
- Check the BT/WLAN feed point
- Check RFU5404 & RFU5403

GPS Function Test Procedure

- Check GPS function with GPS Test Tool.
- Check the GPS spring.
- Check RFU5504 (LNA) and surrounding components
- Check the 1.8V supply voltage of GPS LNA

1.



10mm * 2.5mm
conductive foam

5mm * 2.5mm
conductive foam

10mm * 6.5mm
conductive fabric

TOP side of the motherboard attached gasket,
objective To strengthen the grounding of
the motherboard with front frame Iron parts,
beneficial to suppress the GPS / WIFI / Diversity antenna efficiency
and system integration interference on the motherboard.

Conductive foam thickness of the motherboard
are compressible 1mm to 0.3mm.

This conductive fabric is designed to make the front camera noise to-ground,
and then suppressing noise to the interference of GPS signals

10mm*2.5mmConductive foam: 5pcs

5mm*2.5mmConductive foam: 1pcs

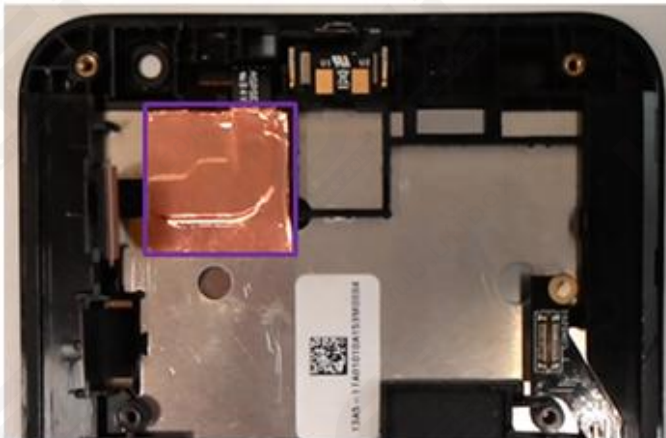
10mm*6.5mmConductive fabric : 1pcs

2.



Front frame iron parts attached gasket, to strengthen the grounding and the front frame iron parts of sub-board, beneficial to the sub-board 2/3 / 4G antenna efficiency and system integration interference suppression

11mm * 3mm conductive foam: 4pce
Irregularly shaped conductive fabric & gasket: 2pce



The purpose of this copper foil attached, to suppress TP FPC noise interference on GPS / WIFI antenna